GRAPHICAL REPRESENTATION OF THE SOLUTION OF TYPES OF EQUATIONS AND SYSTEMS OF EQUATIONS

THESIS

GRAPHICAL REPRESENTATION OF THE SOLUTION OF TYPES OF EQUATIONS AND SYSTEMS OF EQUATIONS

THESIS

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of the Requirements

For the Degree of

MASTER OF ARTS

By

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CHAPTER I

ONE LINEAR EQUATION

In this chapter one line will be considered and therefore we will deal with one linear equation. Practically all of this work will be done upon the general equation.

Take the equation

$$2x + y = 2.$$

This equation is plotted on Fig. 1 which is the graphical representation of it.

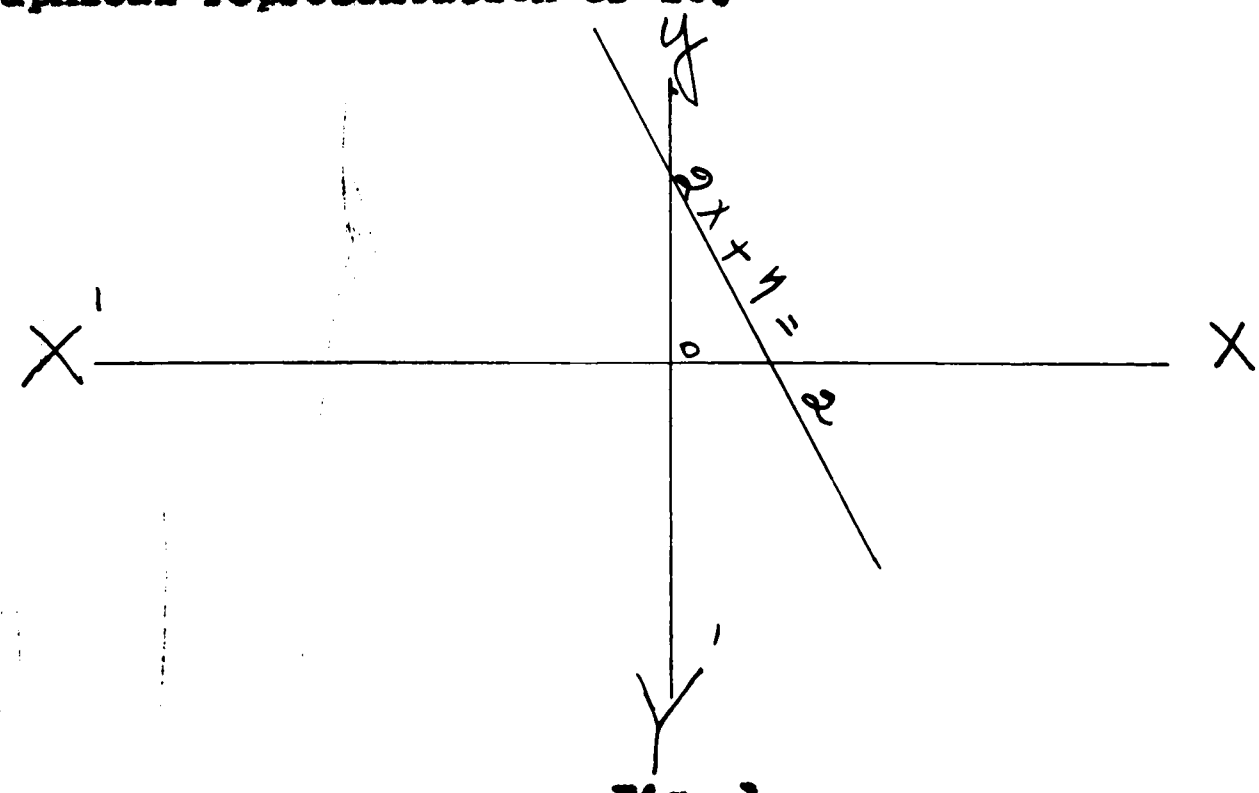


Fig. 1

We will take the general equation $ax + by = c$, of which the above equation is a special form. For purposes of plotting consider here that $a > 0$, $b > 0$,

$c > 0$. The graphical representation of this equation is shown on Fig. 2.

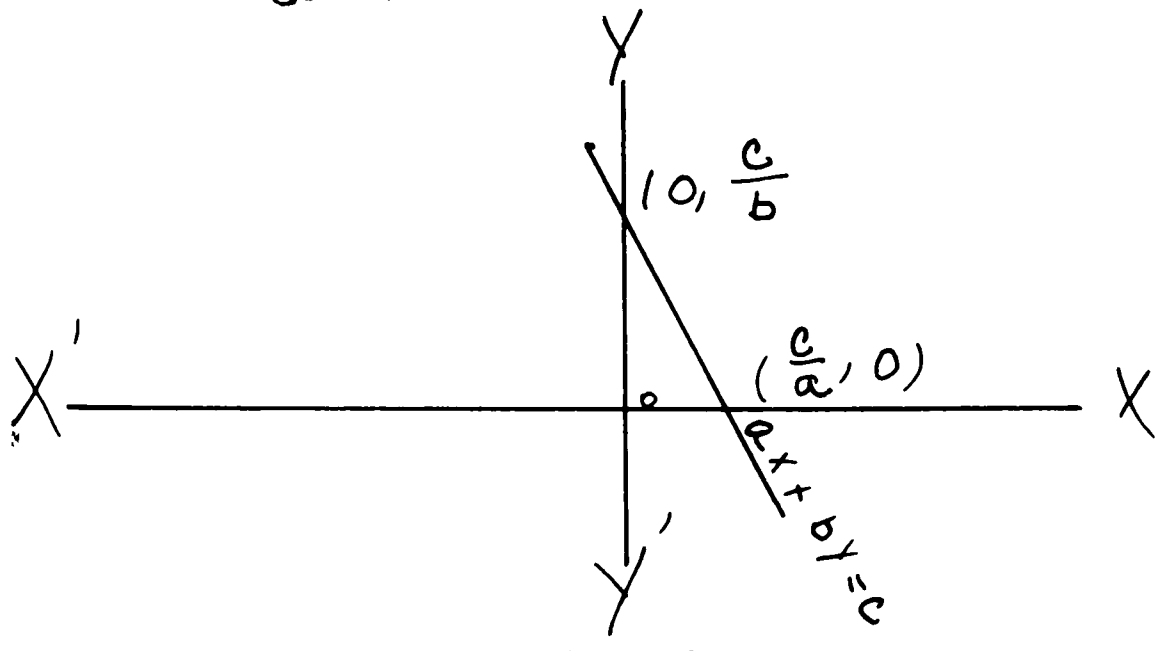


Fig. 2

Assume the following conditions for the equation $ax + by = c$.

If $a = 0$, $b \neq 0$, then $by = c$ or $y = \frac{c}{b}$, the line coincides with the x-axis if $c = 0$ but if $c \neq 0$, it is parallel to the x-axis, above or below according as <u>b</u> and <u>c</u> have like or unlike signs as shown in Fig. 3.

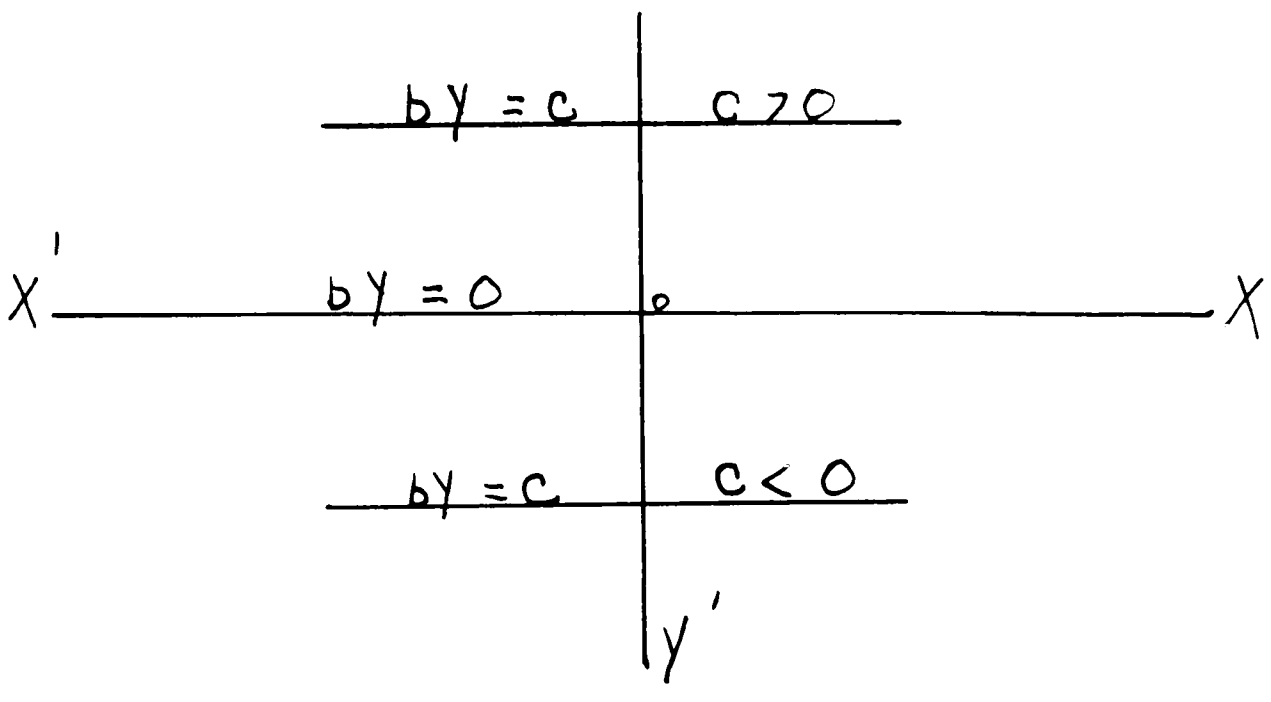


Fig. 3

If $b = 0$, $a \neq 0$, then $ax = c$, or $x = \frac{c}{a}$, the line coincides with the y-axis if $c = 0$ but if $c \neq 0$ it is parallel to the y-axis to the right or left according as a and c have like or unlike signs as shown in Fig. 4.

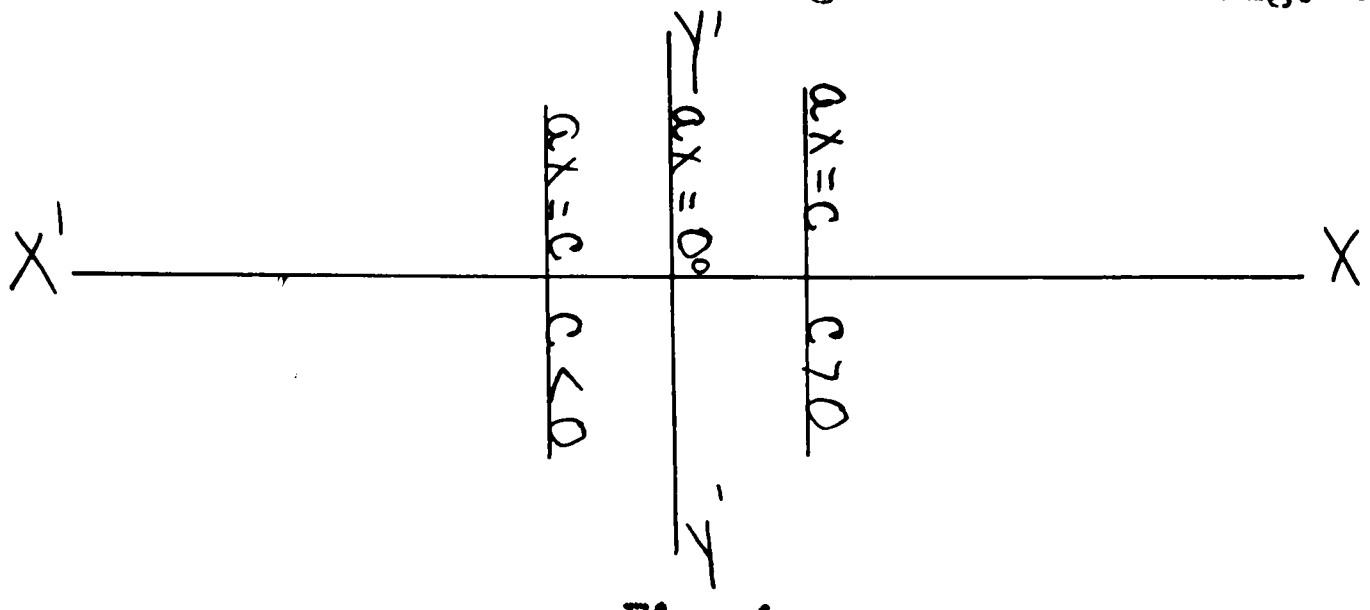


Fig. 4

The case $c = 0$, $a = 0$, $b \neq 0$ and the case $c = 0$, $a \neq 0$, $b = 0$ have been treated just above.

If $c = 0$, $b \neq 0$, $a \neq 0$, then $ax + by = 0$. The line passes through the origin extending through but not out of quadrants II and IV or I and III, according as a and b have like or unlike signs respectively as shown in Fig. 5.

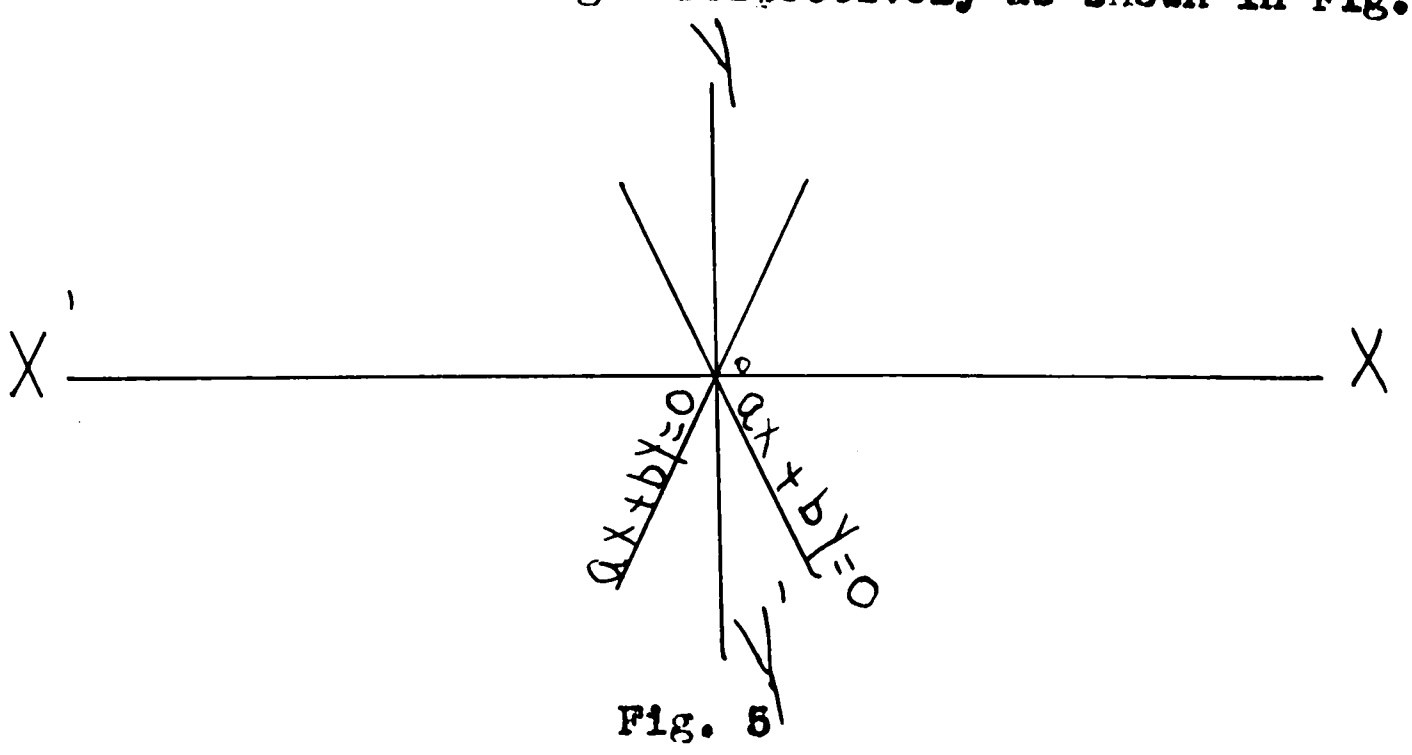


Fig. 5

If a = b = c = 0, the equation does not exist. Also for a = b = 0, the equation does not exist and in these cases no graph is possible.

Again we will assume the following conditions for the equation, ax + by = c.

If all constants have like signs, the line passes completely across the first quadrant as shown in Fig. 2.

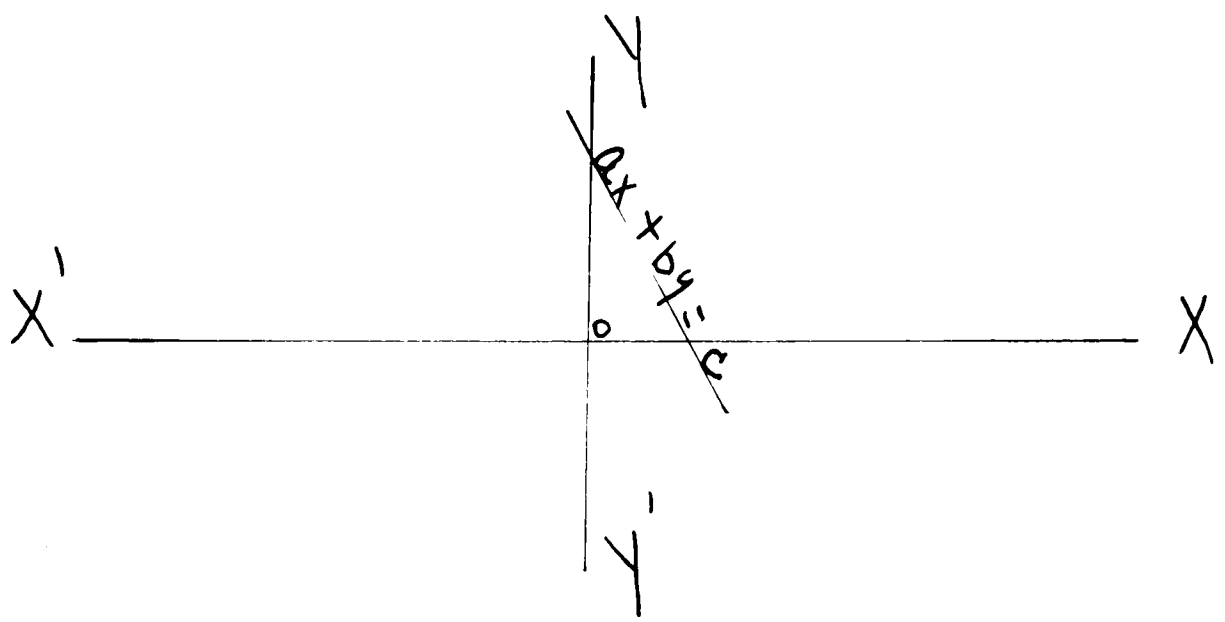


Fig. 2

If a is unlike in sign from both b and c, then the line passes completely across the second quadrant as shown in Fig. 6.

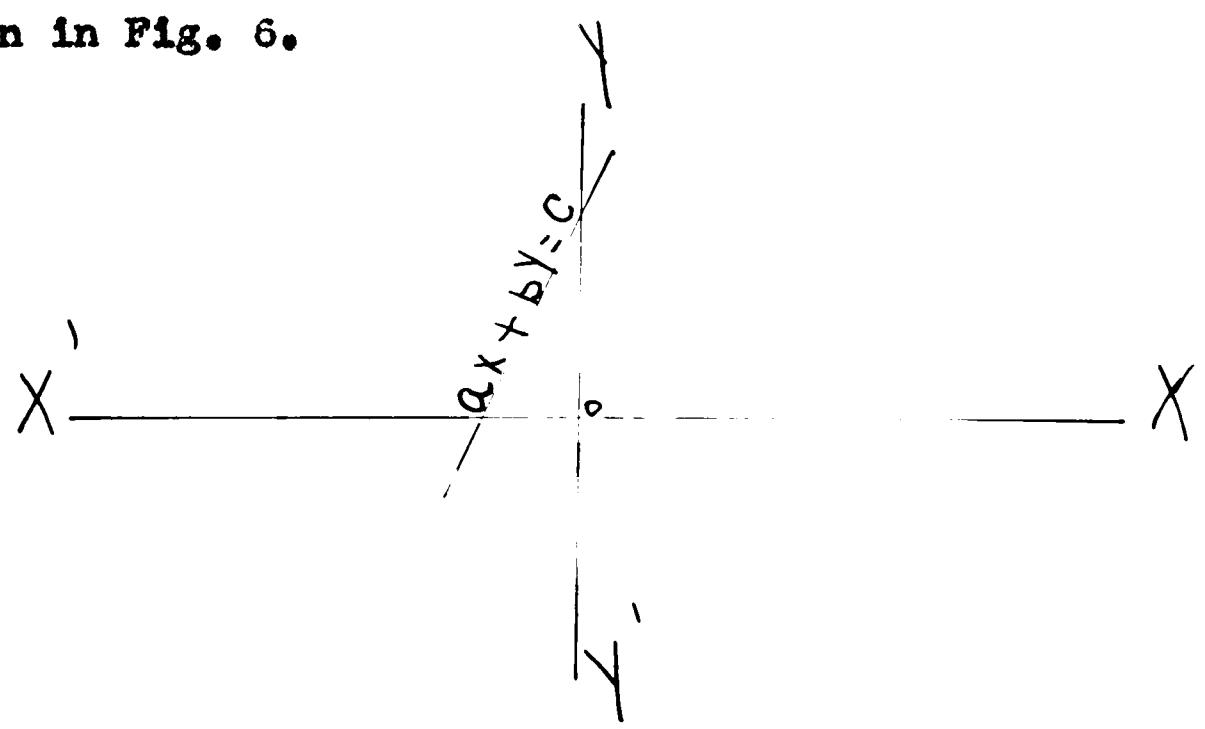


Fig. 6

If b is unlike in sign from both a and c, then the line passes completely across the fourth quadrant as shown in Fig. 7.

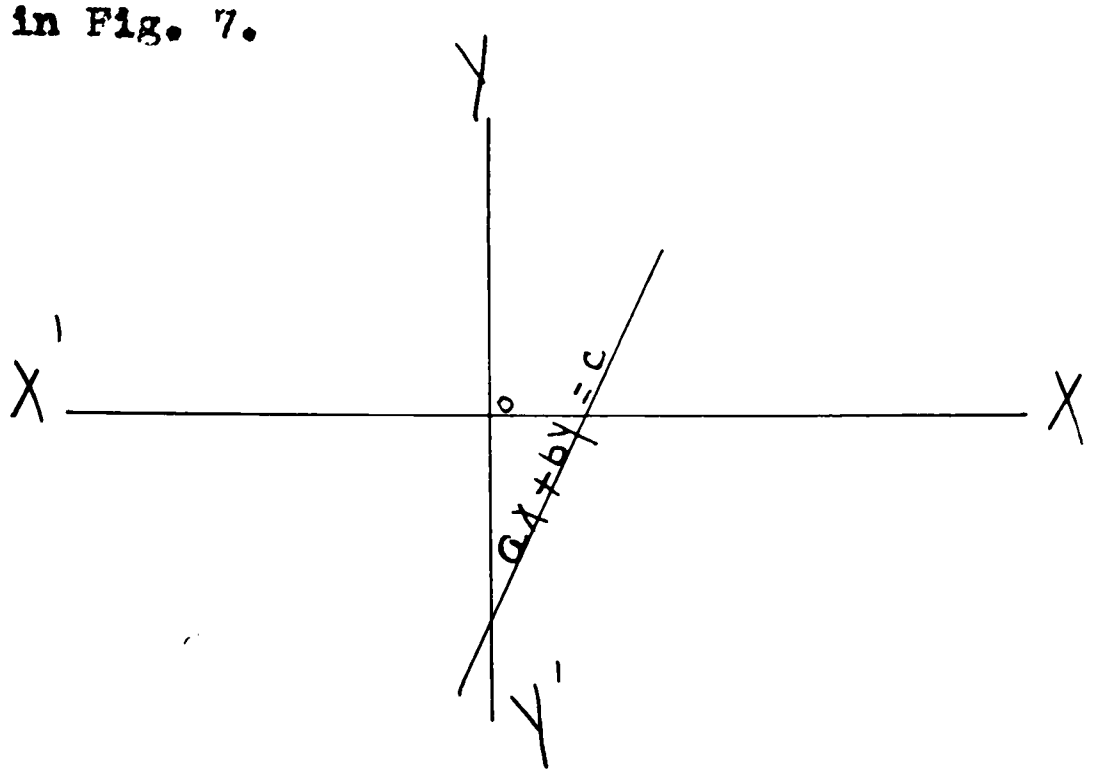


Fig. 7

If c is unlike in sign from both a and b, then the line passes completely across the third quadrant as shown in Fig. 8.

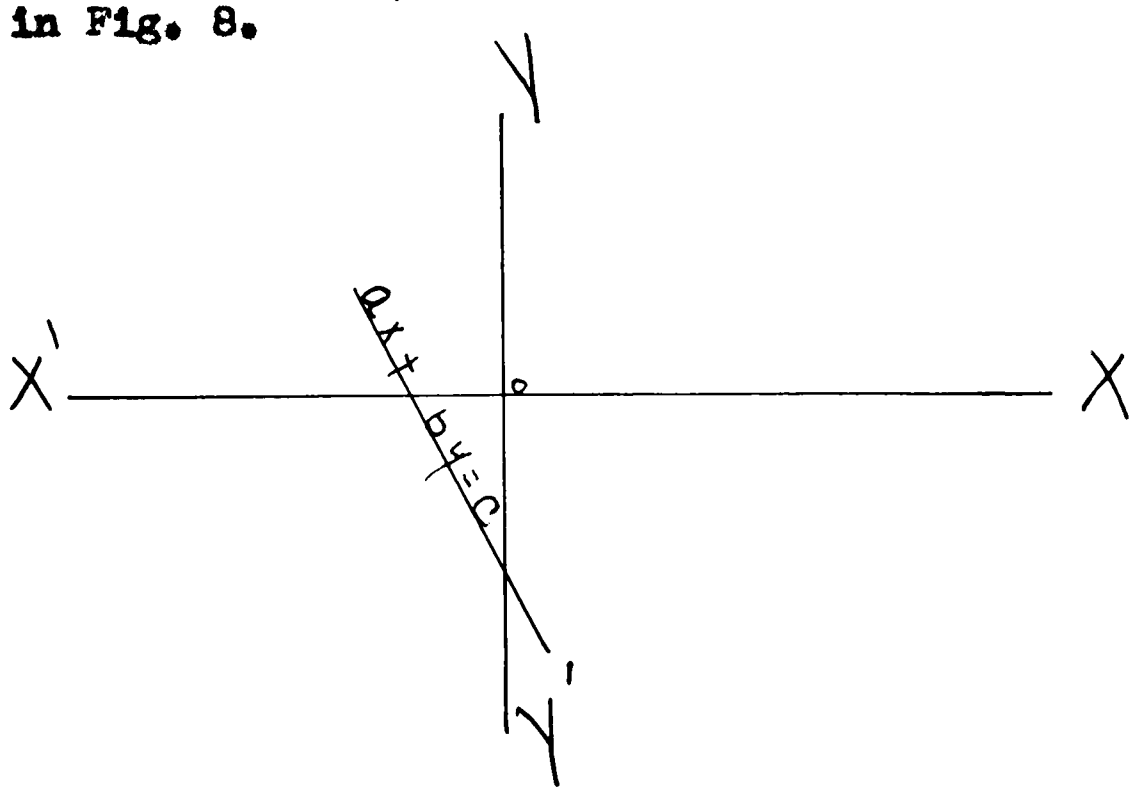


Fig. 8

CHAPTER II

TWO LINEAR EQUATIONS

In this chapter two straight lines are considered and therefore we will use two linear equations. Practically all of this work will be done with the general equations.

Take the equations

$$5x - 2y = 10$$
$$2x + 3y = 6$$

Solving we get,

$$10x - 4y = 20$$
$$10x + 15y = 30$$
$$-19y = -10$$
$$y = \frac{10}{19}$$

$$x = \frac{42}{19}$$

These equations are plotted on Fig. 9 which is

the geometric representation of them.

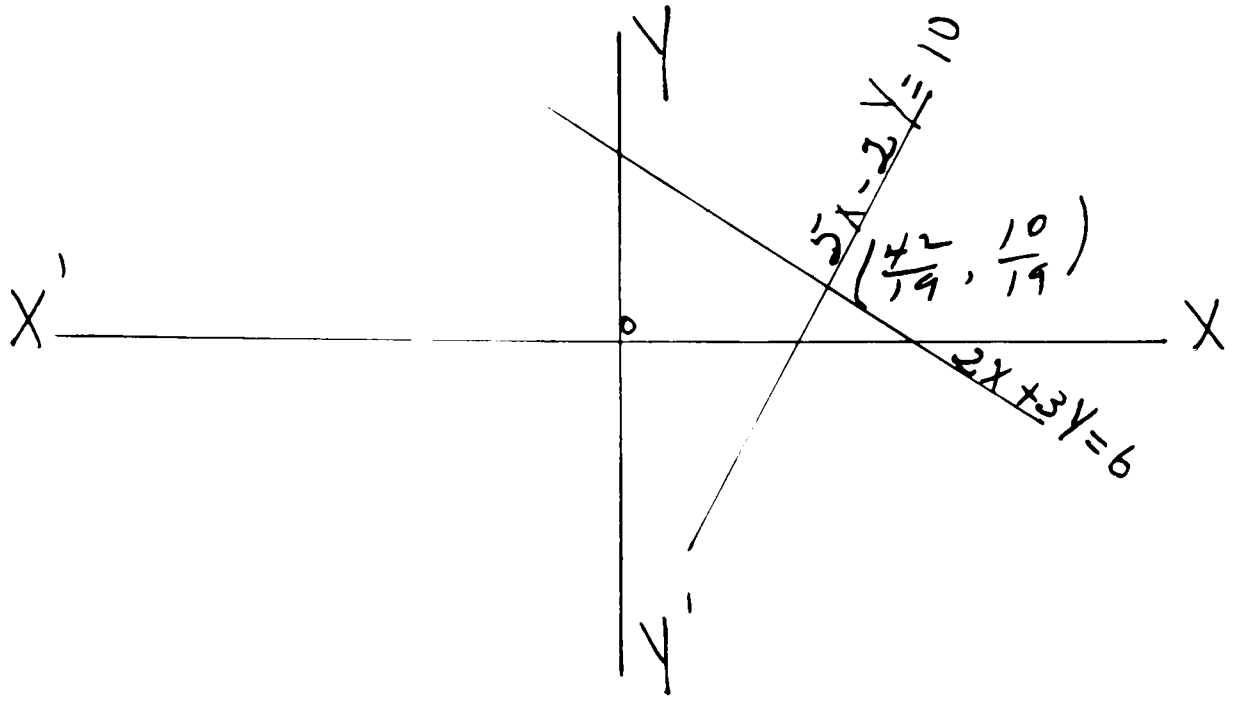


Fig. 9

We will take the general equations of which the above is a special form.

$$(1) \qquad ax + by = c$$

$$(2) \qquad dx + ey = f.$$

We will solve this system of equations algebraically.

$$x = \frac{\begin{vmatrix} c & b \\ f & e \end{vmatrix}}{\begin{vmatrix} a & b \\ d & e \end{vmatrix}} = \frac{ce - bf}{ae - bd}$$

$$y = \frac{\begin{vmatrix} a & c \\ d & f \end{vmatrix}}{\begin{vmatrix} a & b \\ d & e \end{vmatrix}} = \frac{af - cd}{ae - bd}, \quad ae - bd \neq 0.$$

Suppose that $ae - bd \neq 0$ and for the purpose of showing the solution graphically, we will let $a = 1$, $b = 1$, $c = 4$, $d = 2$, $e = -3$ and $f = 3$, then

(1) $ax + by = c$

(2) $dx + ey = f$

becomes

$$x + y = 4$$

$$2x - 3y = 3.$$

Solving the special form we will get $x = 3$ and $y = 1$. The special form is plotted on Fig. 10.

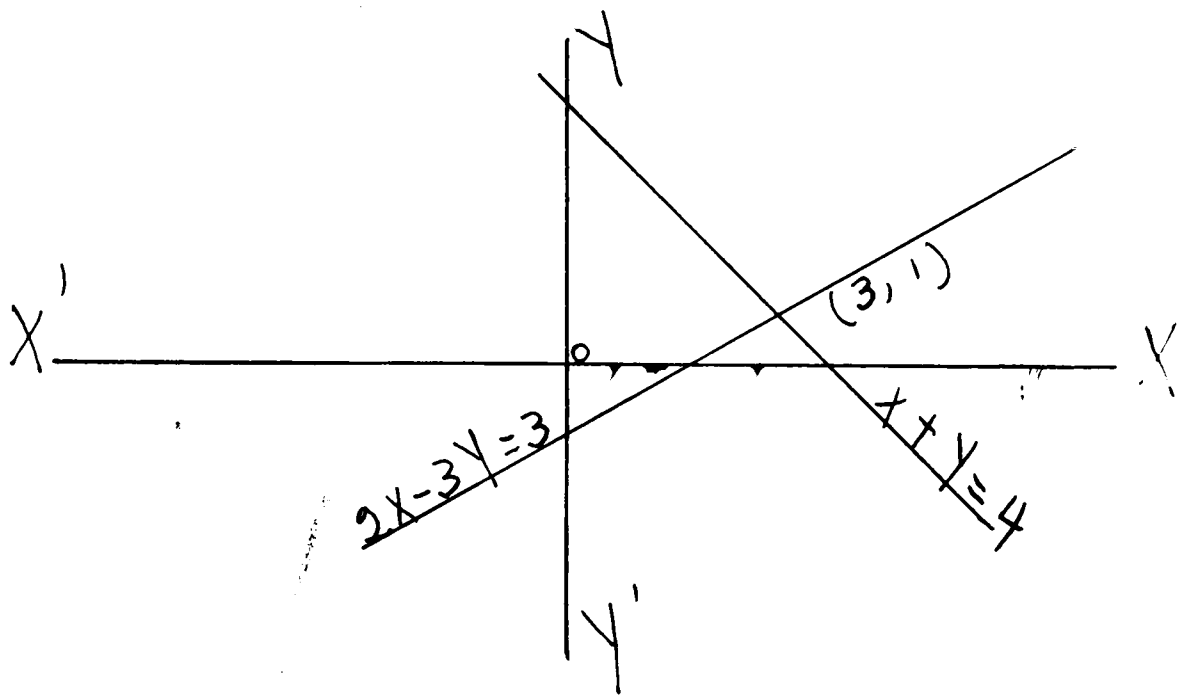


Fig. 10

In (1) if $x = 0$, then $y = \frac{c}{b}$, $b \neq 0$

and if $y = 0$, then $x = \frac{c}{a}$, $a \neq 0$

In (2) if $x = 0$, then $y = \frac{f}{e}$, $e \neq 0$

and if $y = o$, then $x = \frac{f}{d}$, $d \neq 0$.

Fig. 11 contains the plot of the general equations.

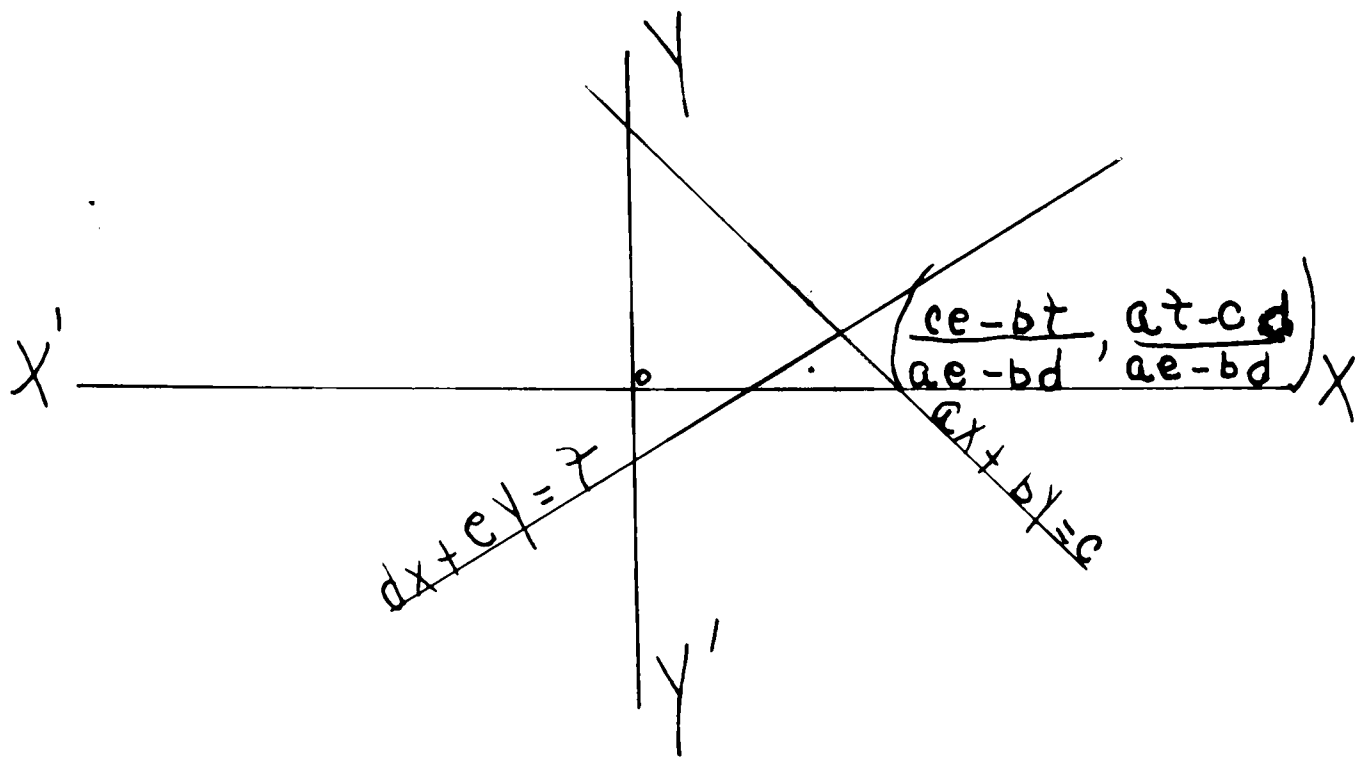


Fig. 11

Suppose that ae- bd = 0, we get ae = bd, or

$$a = \frac{bd}{e}, \quad (e \neq 0).$$

Substituting this value in equation (1)

$$\frac{bdx}{e} + by = c$$

$$dx + ey = f.$$

Simplifying

$$bdx + bey = ce$$

$$dx + ey = f.$$

Dividing (1) by b

$$dx + ey = \frac{ce}{b}$$

$$b \neq 0$$

$$dx + ey = f.$$

It can be observed that these two equations are identical on the left side of the equality signs.

We will assume $x = k_1$, where k_1 is a constant and find the difference between the corresponding y's in the two equations. This length lies parallel to the y-axis, which is shown by the following work.

$$dx + ey = \frac{ce}{b}$$

$$dx + ey = f.$$

In the first of these two equations let $x = k_1$, then

$$dk_1 + ey = \frac{ce}{b}$$

$$ey = \frac{ce}{b} - dk_1.$$

Designate this value of y by y_1,

$$y_1 = \frac{c}{b} - \frac{dk}{e}, \quad e \neq 0.$$

In the second of the above equations let $x = k_1$, then

$$dk_1 + ey = f.$$

Designate this value of y by y_2,

$$y_2 = \frac{f}{e} - \frac{dk^1}{e}$$

$$y_1 - y_2 = \left(\frac{c}{b} - \frac{dk_1}{e} \right) - \left(\frac{f}{e} - \frac{dk_1}{e} \right)$$

$$y_1 - y_2 = \frac{c}{b} - \frac{f}{e} = \frac{ce - bf}{be} .$$

Now we will let x have the value of k_2, ($k_1 \neq k_2$), where k_2 is a constant, and find the difference of the two corresponding y's. This length is parallel to the y-axis.

$$dx + ey = \frac{ce}{b}$$

$$dx + ey = f$$

$$dk_2 + ey = \frac{ce}{b}$$

$$ey = \frac{ce}{b} - dk_2 .$$

Designate this value of y by y_3,

$$y_3 = \frac{c}{b} - \frac{dk_2}{e}$$

$$dk_2 + ey = f$$

$$ey = f - dk_2 .$$

Designate this value of y by y_4,

$$y_4 = \frac{f}{e} - \frac{dk_2}{e}$$

$$y_3 - y_4 = \left(\frac{c}{b} - \frac{dk_2}{e} \right) - \left(\frac{f}{e} - \frac{dk_2}{e} \right)$$

$$y_3 - y_4 = \frac{c}{b} - \frac{f}{e} = \frac{ce - bf}{be} .$$

For the value of x, $x = k_2$ in both equations, $y_1 - y_2$ represents the difference between the corresponding two y's, y_1 and y_2 . This difference $y_1 - y_2 = \frac{c}{b} - \frac{f}{e}$ is the same as the difference of $y_3 - y_4 = \frac{c}{b} - \frac{f}{e}$, the value

of y, y_3, y_4 for the second value of x, $x = k_2$ used in both equations simultaneously.

The system of equations,

$$dx + ey = \frac{ce}{b}$$

$$dx + ey = f,$$

are plotted and solved graphically on Fig. 12. AB and CD are both parallel to y-axis and are therefore parallel to each other. The figure ABCD is a parallelogram, since one pair of opposite sides AB and CD are parallel and equal. Therefore BC and AD are parallel, that is the lines $dx + ey = \frac{ce}{b}$ and $dx + ey = f$ are parallel being opposite sides of a parallelogram.

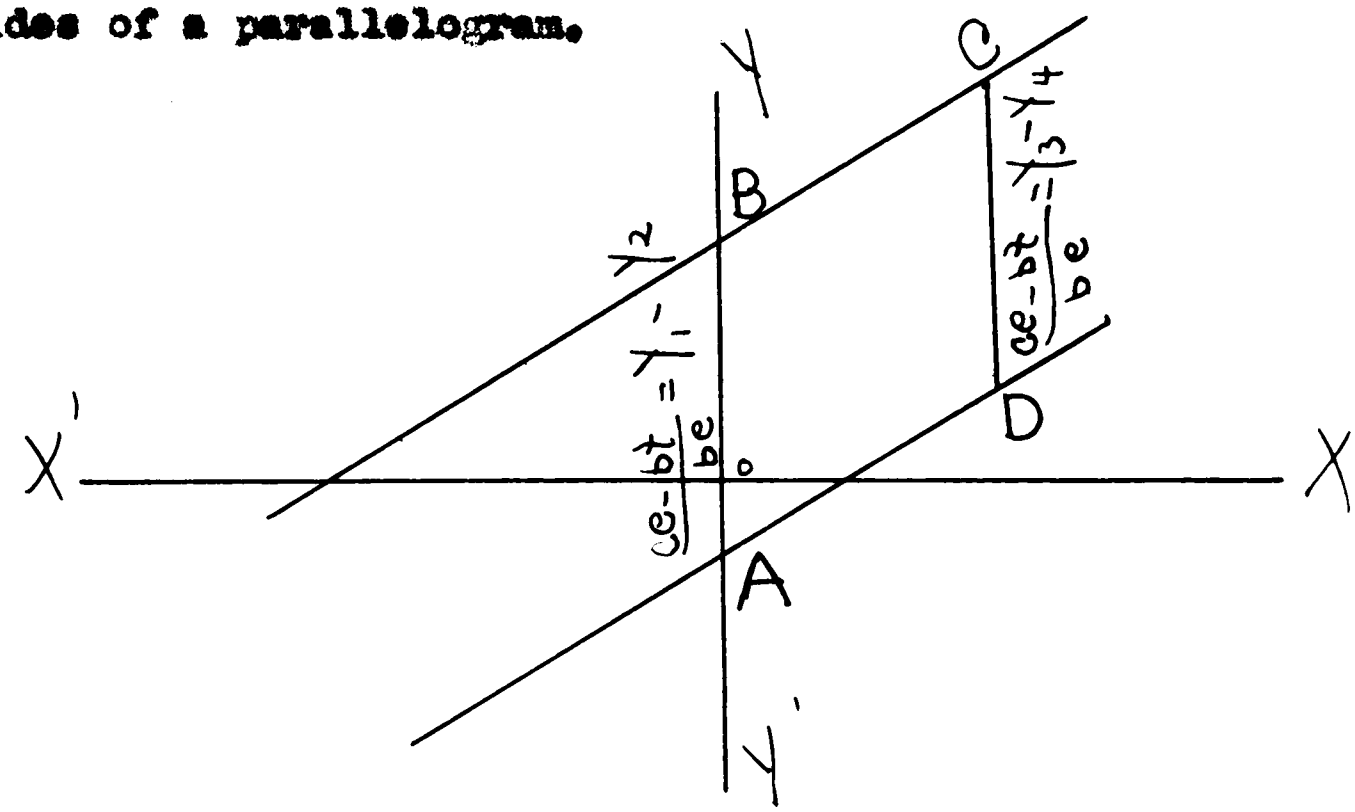


Fig. 12

In the system of equations

(1) $ax + by = c$

(2) $dx + ey = f$

assume $a = 0$ in (1), then $y = \frac{c}{b}$, and substituting this value of y in (2), $x = \frac{bf - ce}{bd}$. This line is parallel to x-axis intersecting the other line at $\left(\frac{bf - ce}{bd}, \frac{c}{b}\right)$ where $bd \neq 0$, Fig. 13.

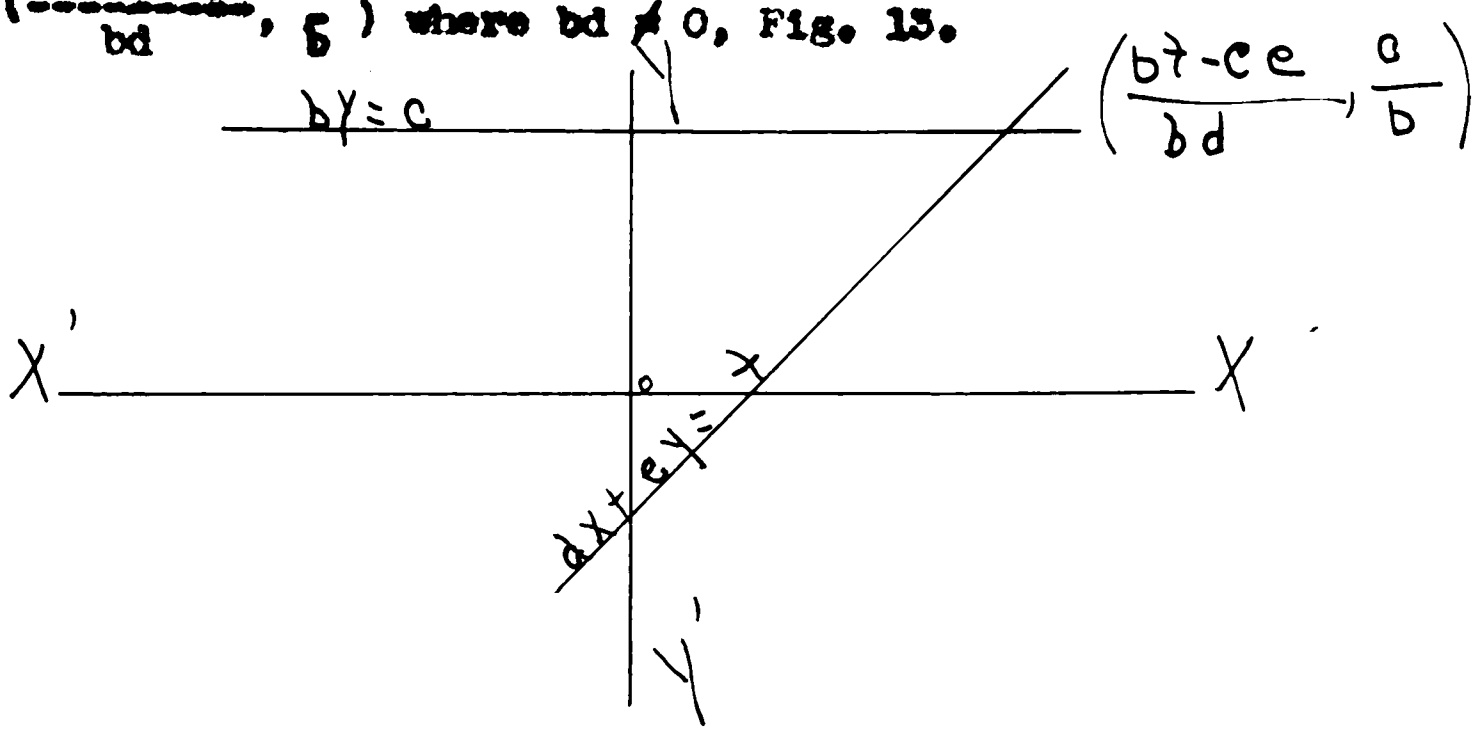


Fig. 13

Assume $b = c$ in (1), then $x = \frac{c}{a}$, substituting this value of x in (2), $y = \frac{af - cd}{ae}$. This line is parallel to y-axis, intersecting the other line at $\left(\frac{c}{a}, \frac{af - cd}{ae}\right)$, where $ae \neq 0$, Fig. 14.

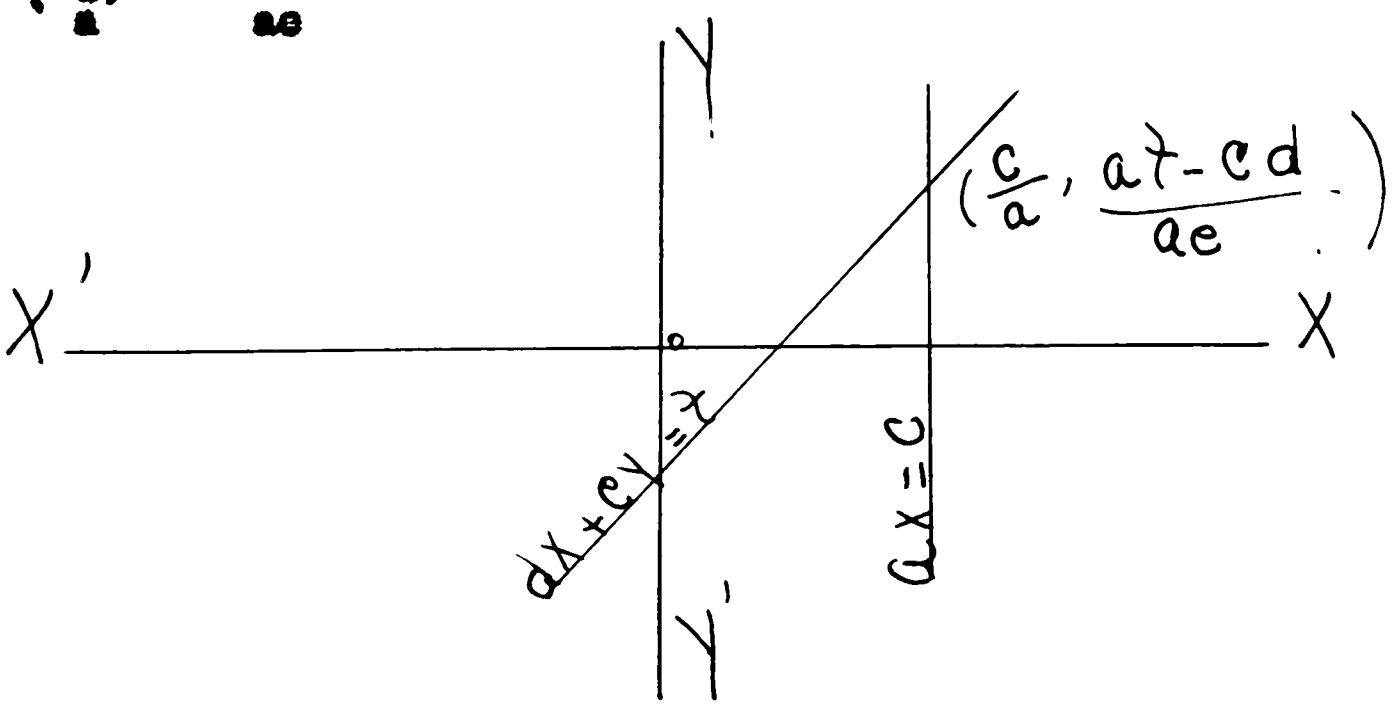


Fig. 14

Assume $c = 0$ in (1), then $ax + by = 0$, or $y = \frac{a}{b}x$. Substituting this value of y in (2), $x = \frac{-bf}{ae - bd}$, then $y = \frac{af}{ae - bd}$. This line passes through the origin intersecting the other line at $\left(\frac{-bf}{ae - bd}, \frac{af}{ae - bd} \right)$ where $ae - bd \neq 0$, Fig. 15.

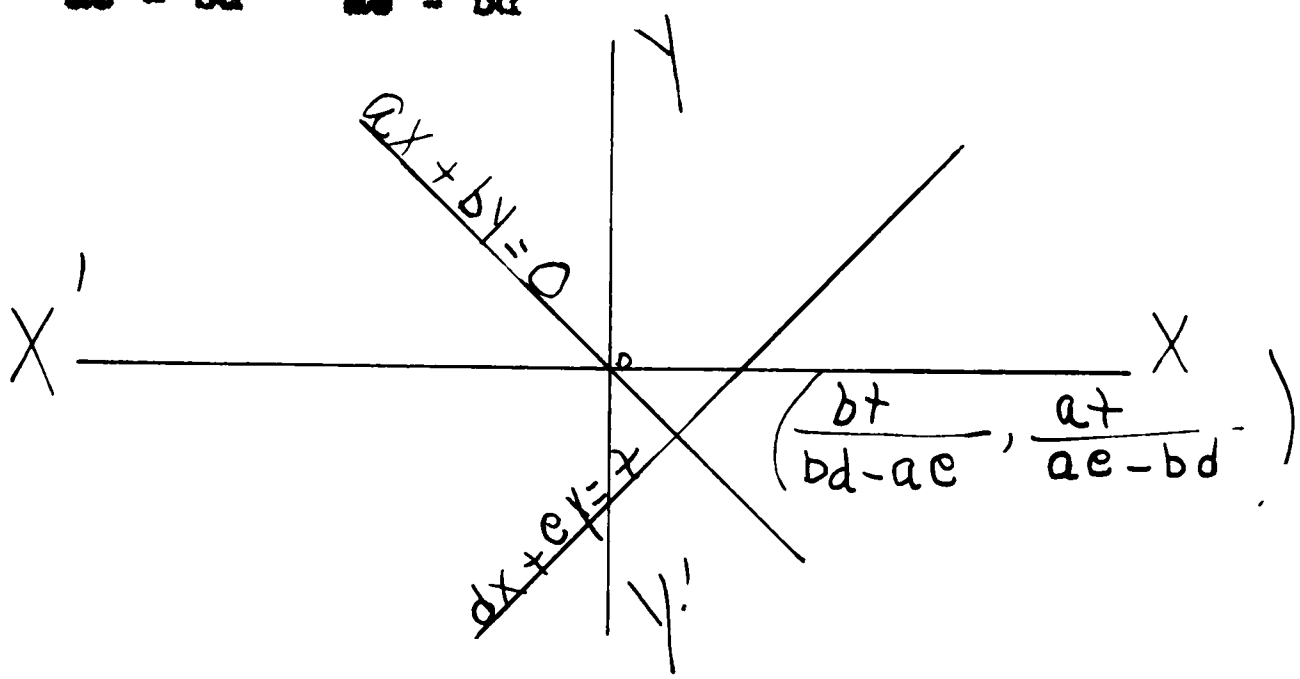


Fig. 15

Assume $d = 0$, then $y = \frac{f}{e}$, and substituting this value of y in (1), $x = \frac{ce - bf}{ae}$. This line is parallel to x-axis intersecting the other line at $\left(\frac{ce - bf}{ae}, \frac{f}{e} \right)$ where $ae \neq 0$, $e \neq 0$ in Fig. 16.

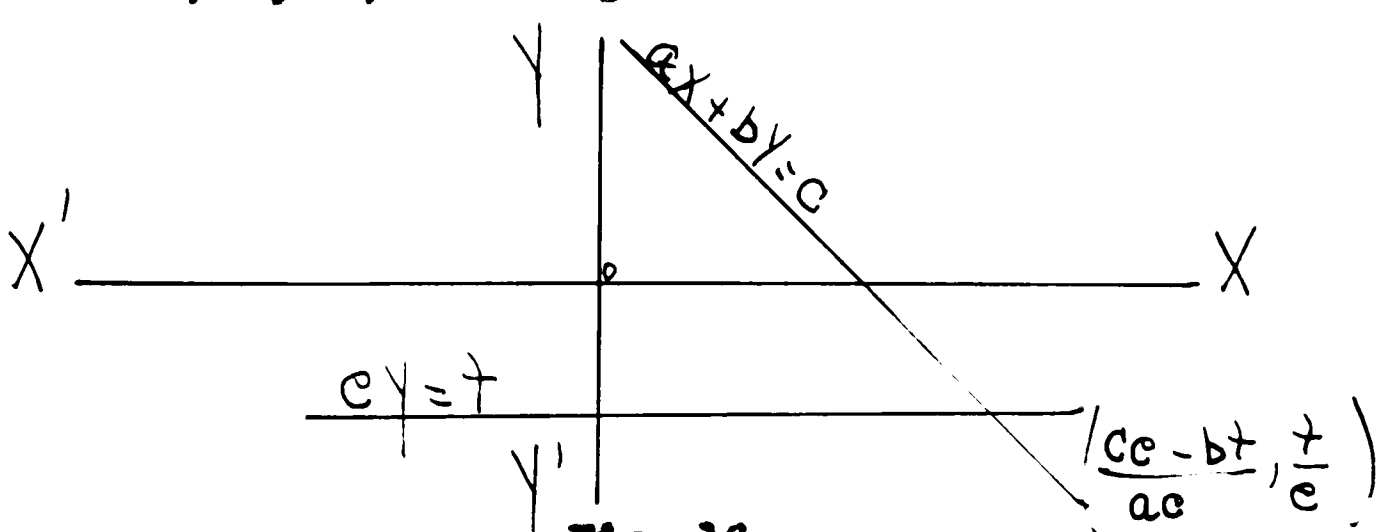


Fig. 16

Assume $e = 0$, then $x = \frac{f}{d}$, and substituting this value of x in (1), $y = \frac{ed - af}{bd}$. This line is parallel to the y-axis intersecting the other line at $\left(\frac{f}{d}, \frac{ed - af}{bd}\right)$ where $bd \neq 0$, Fig. 17.

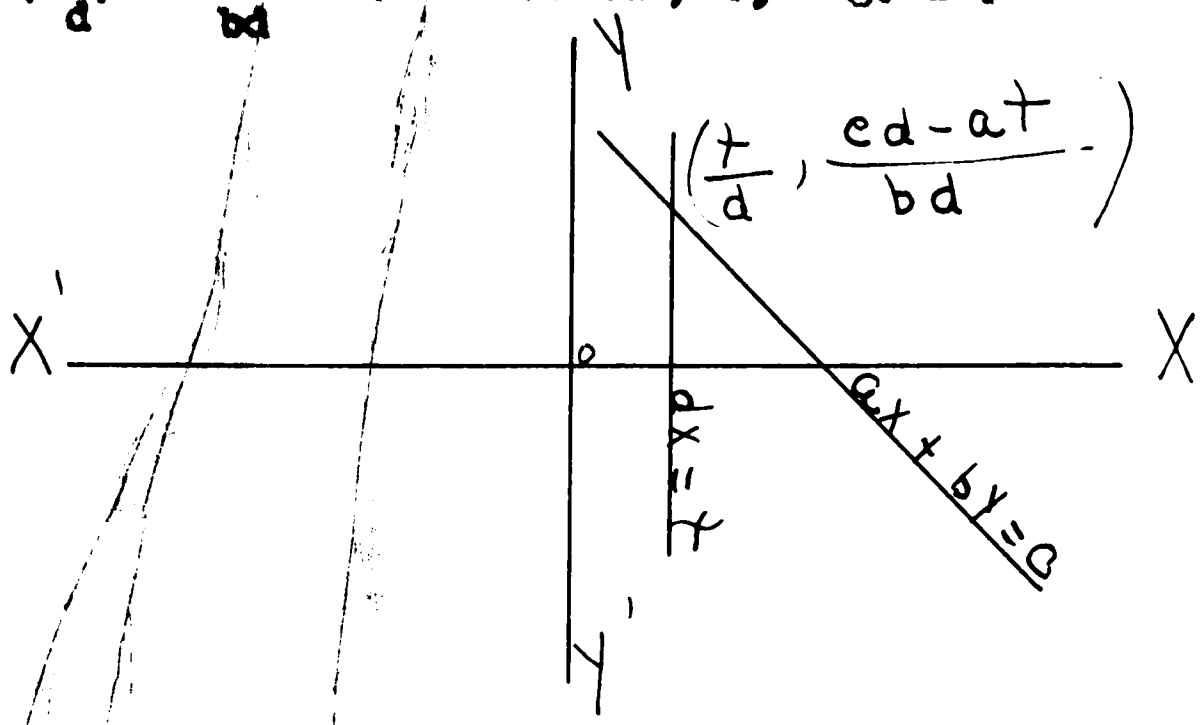


Fig. 17

Assume $f = 0$, then $dx + ey = 0$, or $y = \frac{d}{ex}$.

Substituting this value of y in (1), $x = \frac{-ce}{ae - bd}$, then $y = \frac{de}{ae - bd}$. This line passes through the origin intersecting the other line at $\left(\frac{ce}{ae - bd}, \frac{-de}{ae - bd}\right)$ where $ae - bd \neq 0$, Fig. 18.

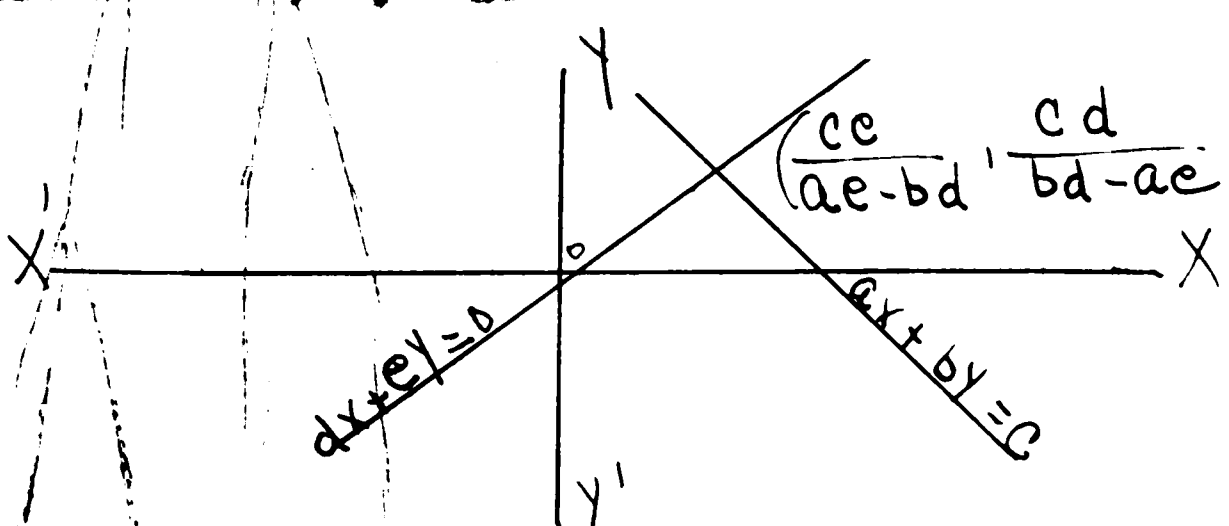


Fig. 18

Assume $a = 0$ and $d = 0$ in (1) and (2) then $y = \frac{c}{b}$, $y = \frac{f}{e}$. The two lines are parallel to the x-axis and comes under the general case, Fig. 19, $ae - bd = 0$.
there is not any common solutions.

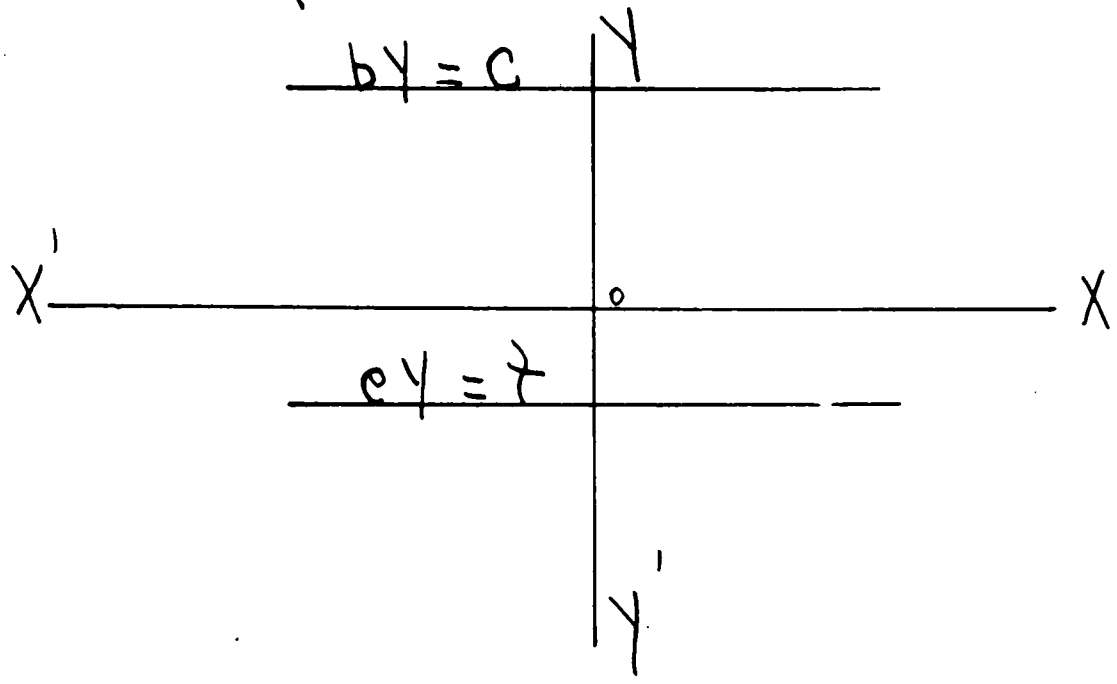


Fig. 19

Assume $a = 0$ and $e = 0$, in (1) $y = \frac{c}{b}$, in (2) $x = \frac{f}{d}$, then line (1) is parallel to x-axis and line (2) is parallel to y-axis intersecting at $\left(\frac{f}{d}, \frac{c}{b}\right)$, where $b \neq 0$, $d \neq 0$, Fig. 20.

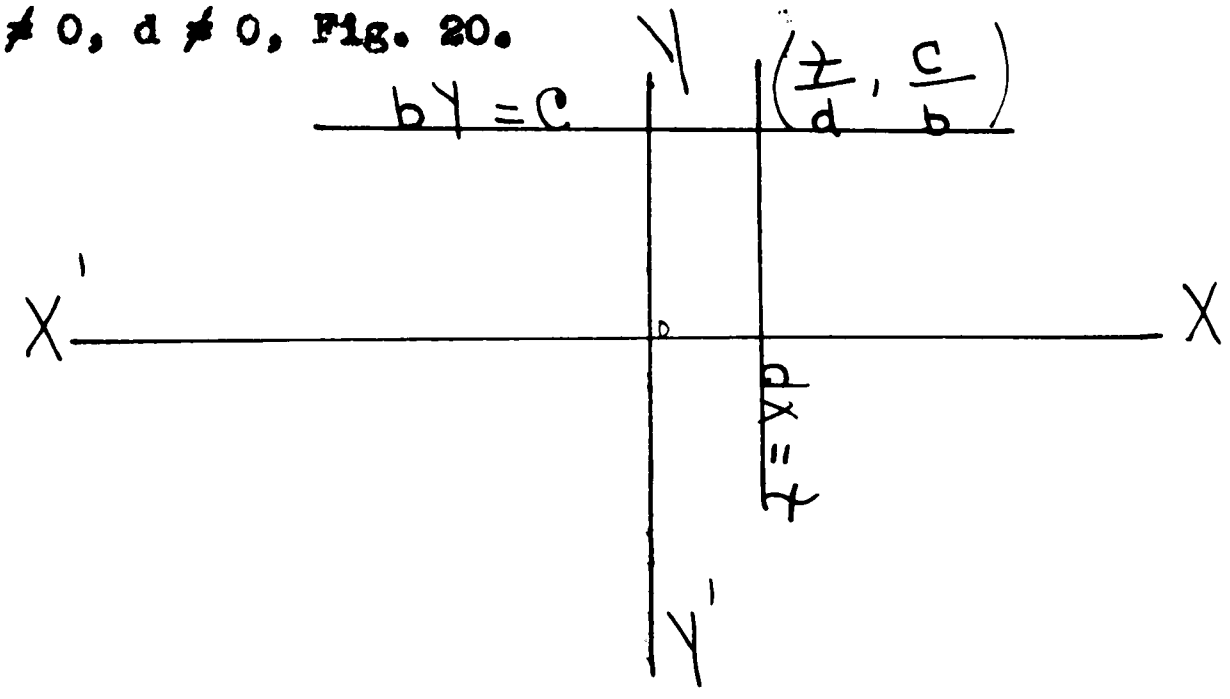


Fig. 20.

Assume $a = 0$, $f = 0$, then in (1) $y = \frac{c}{b}$, in (2) $y = -\frac{d}{e}x$. Line (1) is parallel to x-axis and line (2) passes through the origin, intersecting at $\left(-\frac{ce}{bd}, \frac{c}{b}\right)$ there $bd \neq 0$. Fig. 21.

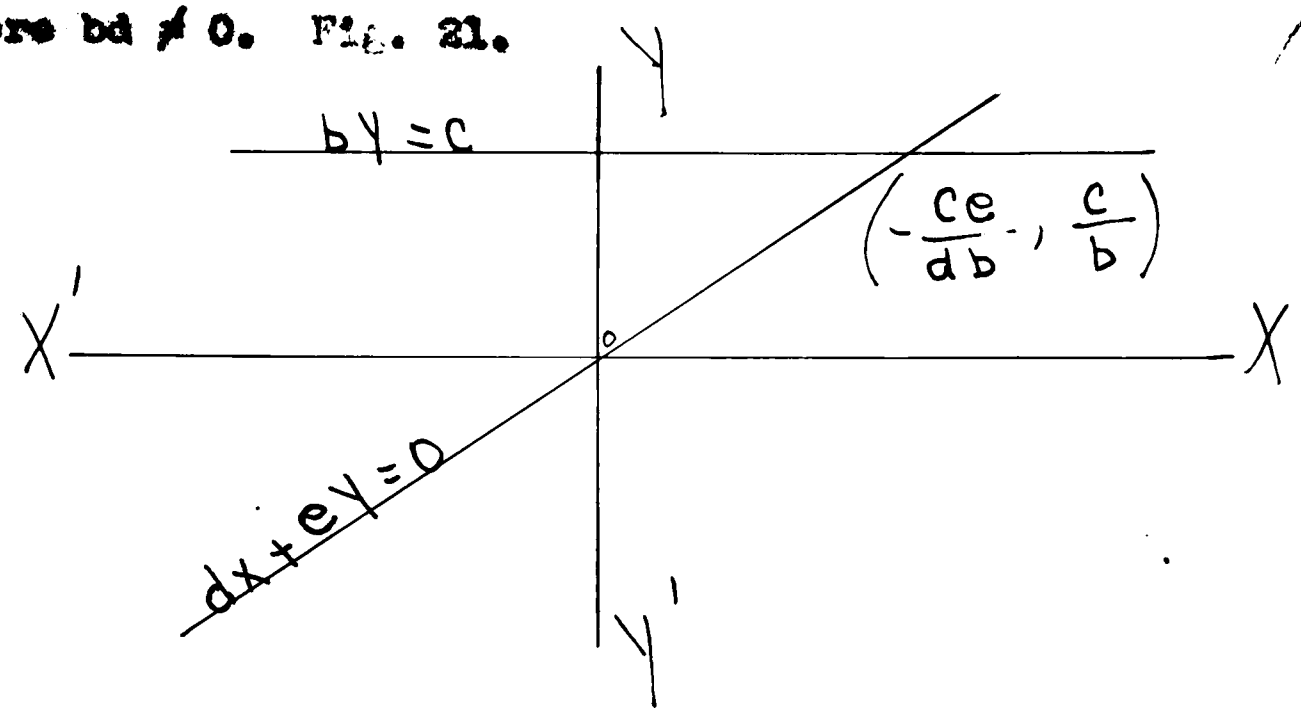


Fig. 21

Assume $b = 0$, $d = 0$, then in (1) $x = \frac{c}{a}$, in (2) $y = \frac{f}{e}$. Line (1) is parallel to y-axis and line (2) is parallel to x-axis, intersecting at $\left(\frac{c}{a}, \frac{f}{e}\right)$, where a or e not equal to zero. Fig. 22.

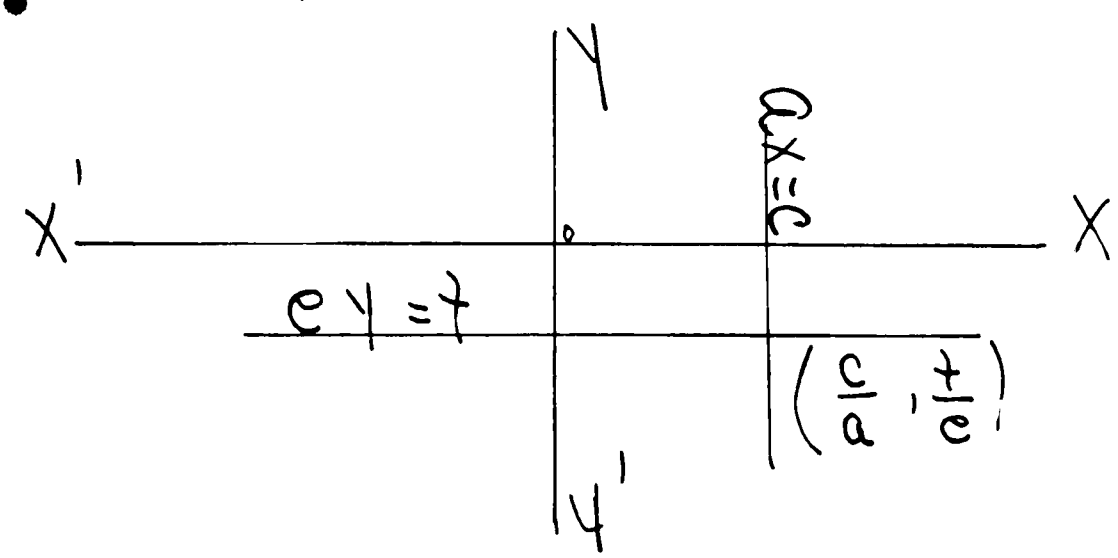


Fig. 22

Assume $b = 0$, $e = 0$, then in (1) $x = \frac{c}{a}$, in (2) $x = \frac{f}{d}$. The two lines are parallel to y-axis and come under the general case $ae - bd = 0$, Fig. 23.
there is not any common solution.

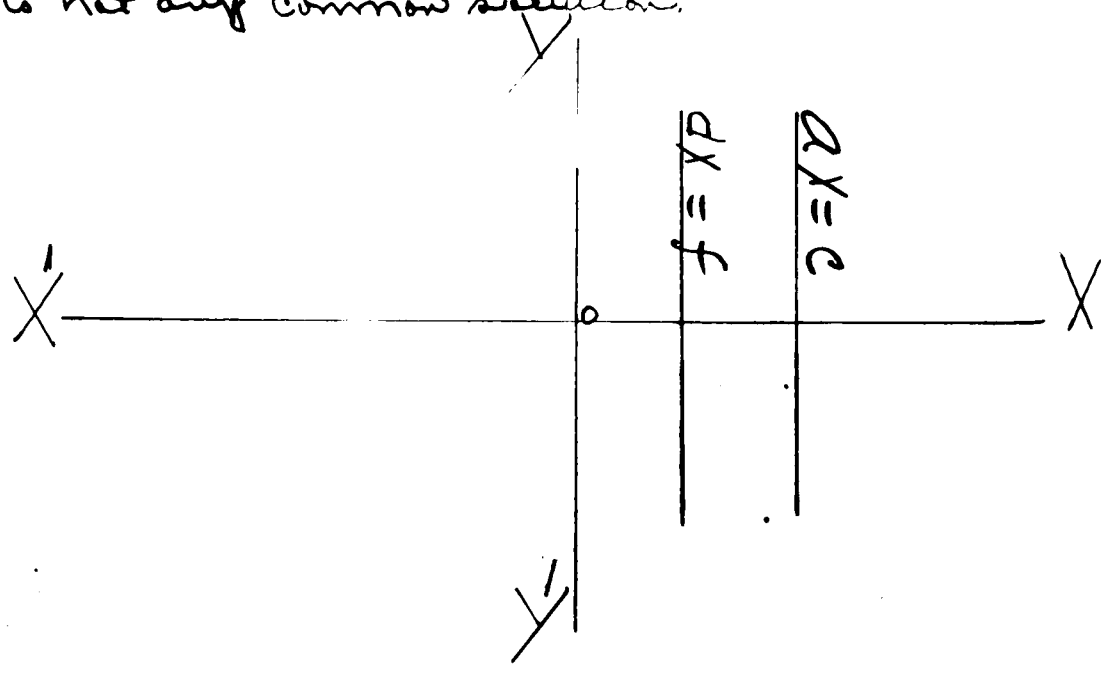


Fig. 23

Assume $f = 0$, $c = 0$, then in (1) $y = -\frac{a}{b}x$, in (2) $y = -\frac{d}{e}x$. The two lines will pass through the origin, intersecting at the origin, Fig. 24.

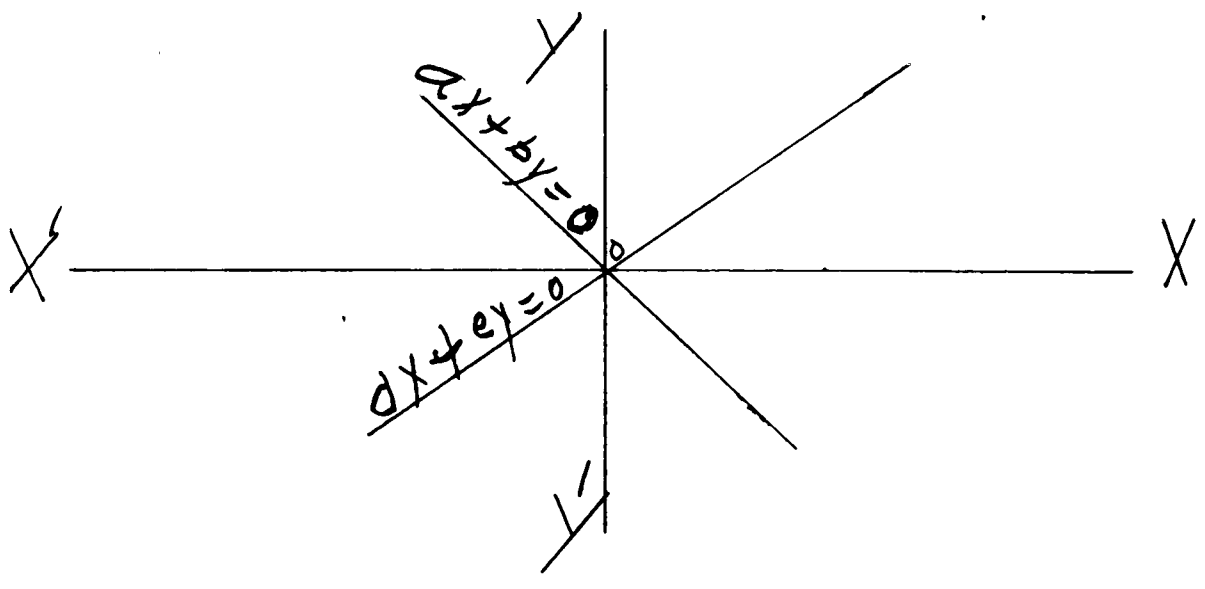


Fig. 24

The coefficients a and b cannot both be zero, similarly d and e cannot be all zero and similarly a, b and c, also d, e and f cannot be all zero.

Assume the condition

$$\frac{a}{d} = \frac{b}{e} = \frac{c}{f}$$

then

$$1.\ \frac{a}{d} = \frac{b}{c}$$

$$2.\ \frac{a}{d} = \frac{c}{f}$$

$$3.\ \frac{b}{e} = \frac{c}{f}.$$

In original case above

$$\frac{a}{d} = \frac{b}{e} = \frac{c}{f}$$

(1) $$\frac{a}{d} = \frac{b}{e}$$

$$a = \frac{bd}{e}$$

(2) $$\frac{a}{d} = \frac{c}{f}$$

$$a = \frac{cd}{f}$$

$$bd = \frac{cd}{f}$$

$\frac{b}{e} = \frac{c}{f}$, which is the same as the third set. Therefore we need to use only two sets of

these equations.

From the first set $\frac{a}{d} = \frac{b}{e}$, we have $a = \frac{bd}{e}$ and

from the third set we find the following,

$$\frac{b}{e} = \frac{c}{f}$$

$$c = \frac{bf}{e}$$

Substituting the value of a and c into equations (1) and (2) of original set, we get,

$$\begin{cases} ax + by = c \\ dx + ey = f \end{cases}$$

$$\begin{cases} \frac{bd}{e} + by = \frac{bf}{e} \\ dx + ey = f \end{cases}$$

$$\begin{cases} bd + bey = bf \\ dx + ey = f \end{cases}$$

$$\begin{cases} dx + ey = f \\ dx + ey = f \end{cases}$$

The equations become identical and the two lines in Fig. 11 coincide as exhibited by Fig. 25.

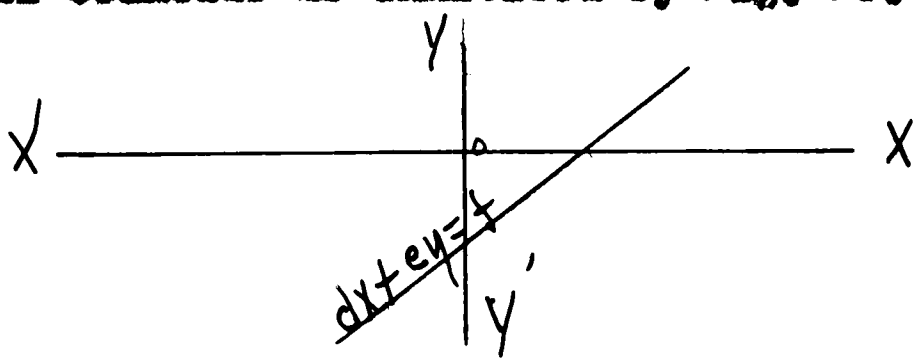
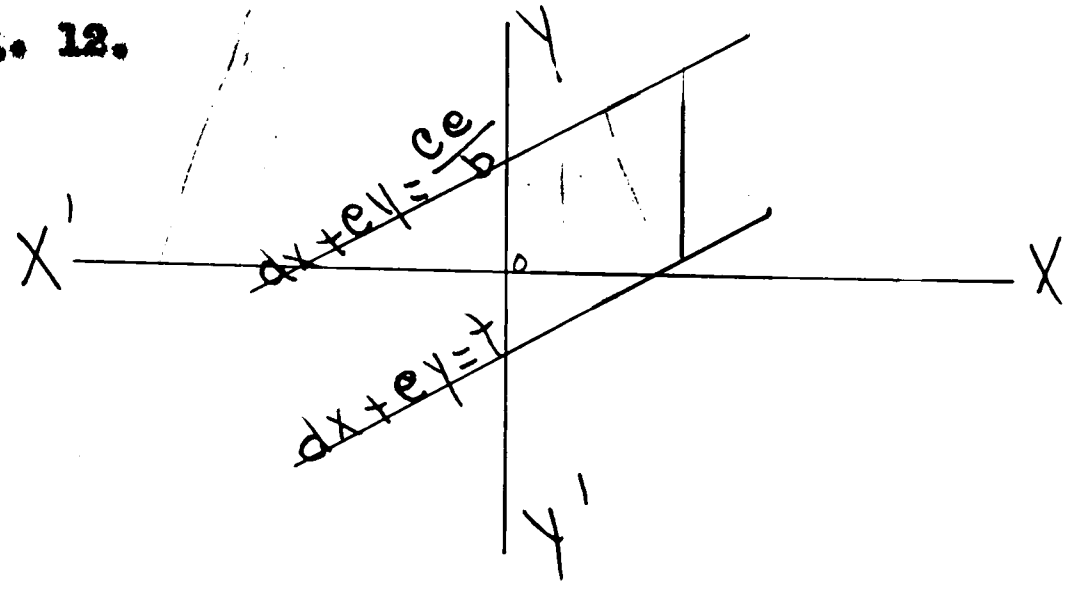


Fig. 25

Assume the condition

$$\frac{a}{d} = \frac{b}{e} \neq \frac{c}{f} .$$

This is treated under ae - bd = 0, shown graphically in Fig. 12.



Assume the condition

$$\frac{a}{d} = \frac{c}{f} = \frac{b}{e}$$

$$a = \frac{cd}{f} .$$

Substitute this value of a in equation (1).

$$\left\{ \frac{cd}{f} x + by = c \right. .$$

$$\left\{ dx + ey = f. \right.$$

$$\left\{ cdx + bfy = cf \right.$$

$$\left\{ dx + ey = f. \right.$$

$$\left\{ dx - f = - \frac{bfy}{e} \right.$$

$$\left\{ dx - f = -ey \right.$$

Equating right members,

$$\frac{bfy}{e} = ey$$

$$\frac{bfy}{e} - ey = 0$$

$$y\,(\,bf - ce\,) = 0$$

$$y = \frac{0}{bf - ce} = 0$$

then $\quad x = \frac{f}{d}.$

This is exhibited graphically in Fig. 26, with the two lines meeting at ($\frac{f}{d}$, 0).

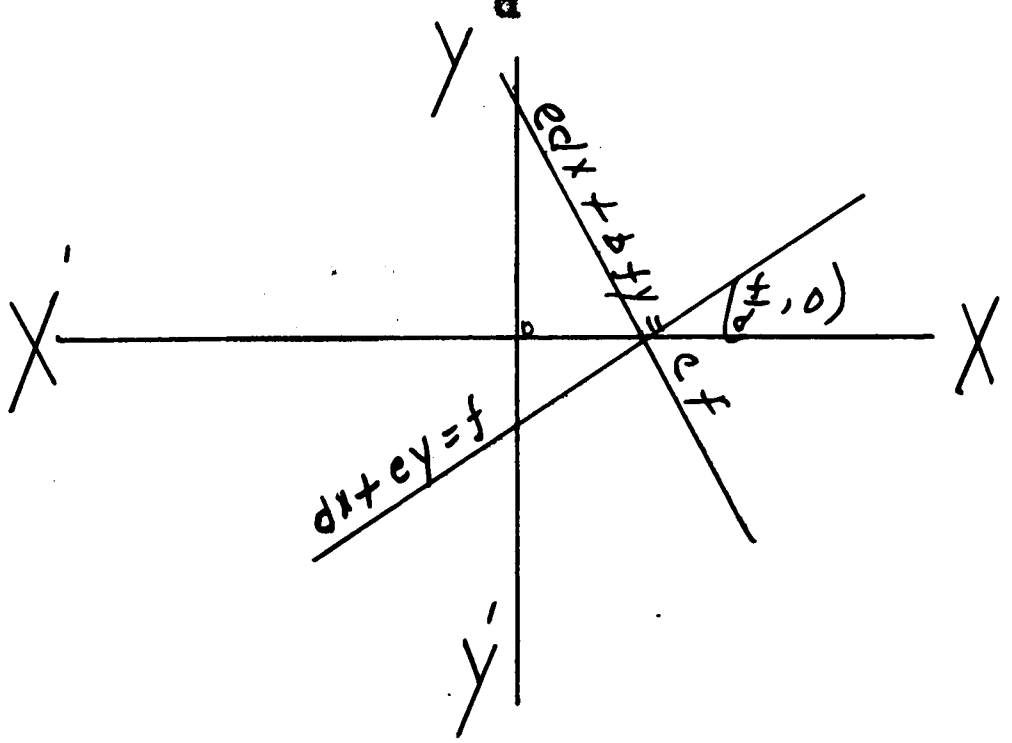


Fig. 26

Assume the condition

$$\frac{b}{a} = \frac{c}{d}$$

$$a = \frac{b}{c}$$

$$b = \frac{ce}{d}$$

Substituting this value for b in the original equation (1).

$$(ax + \frac{ce}{f} y = c$$

$$(dx + ey = f$$

$$(cey - cf = -afx$$

$$(ey - f = -dx$$

$$-afx = -dx$$

$$afx - dx = 0$$

$$x (af - d) = 0$$

$$x = \frac{0}{af - d} = 0$$

then

$$y = \frac{f}{e}.$$

This is exhibited graphically in Fig. 27 with the two lines meeting at $(0, \frac{f}{e})$.

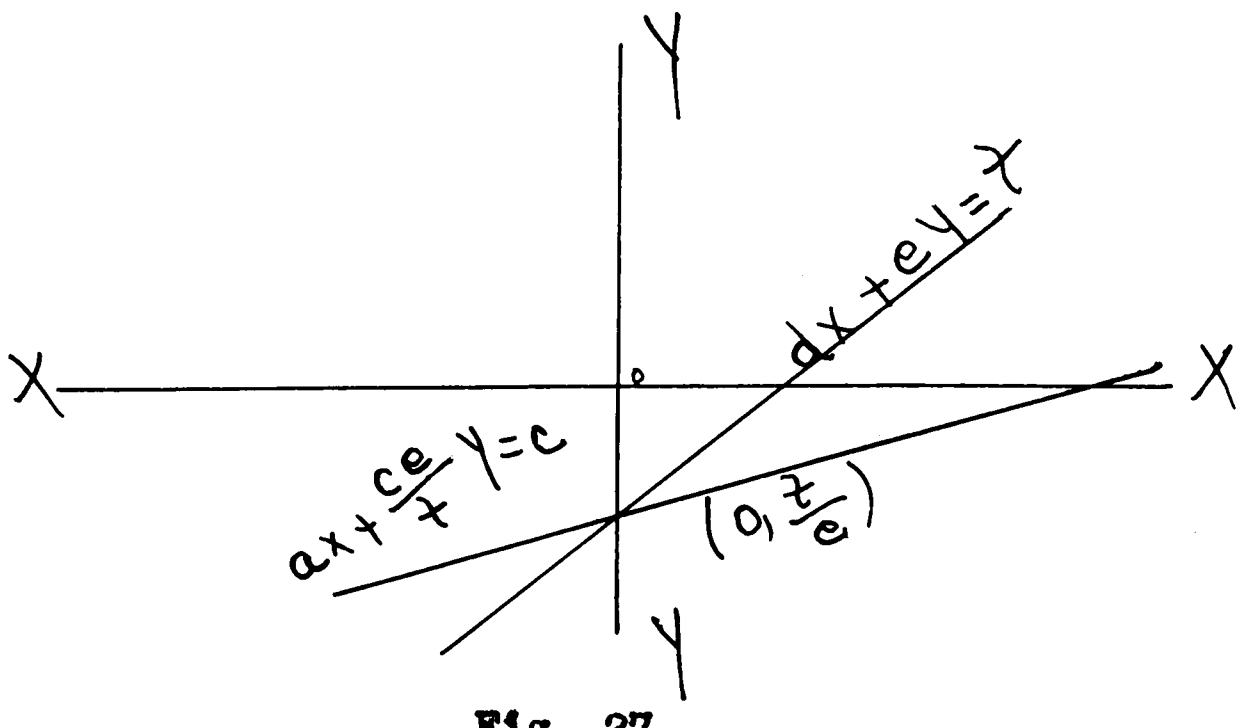


Fig. 27

CHAPTER III

TWO EQUATIONS, ONE OR BOTH NOT LINEAR

This chapter will deal with equations of different degrees with more emphasis upon second degree equations.

We will consider the system of equations.

$$(1) \qquad 2x + y = 4$$

$$(2) \qquad x^2 + y^2 = 4.$$

Solving (1) for y and substituting in (2)

$$(3) \qquad 5x^2 - 16x + 12 = 0$$

$$x = 2, \qquad y = 0$$

$$x = 6/5 \qquad y = 8/5.$$

In Fig. 23 the graphical solution of (1) and (2) is shown with the graph of the equation (3) used for the solution for x. The graph of equation (3) cuts the x-axis at the value of x corresponding to the x values for the common solution of the two original equations, that is at $x = 6/5$ and $x = 2$, and the corresponding values for y in the case just mentioned are obtained by the use of these values of x in equation (1), giving the common

solutions $x = {}^{6}/_{5}$, $y = {}^{8}/_{5}$ and $x = 2$, $y = 0$.

We will consider the equation (1) and (2) as before but getting the auxiliary equation in terms of y.

$$(1) \qquad 2x + y = 4$$

$$(2) \qquad x^2 + y^2 = 4.$$

Solving (1) for x and substituting in (2)

$$(4) \qquad 5y^2 - 8y = 0$$

$$y = 0, \; x = 2$$

$$y = {}^{8}/_{5}, \; x = {}^{6}/_{5}.$$

The graphical representation is shown in Fig. 28. The graph of equation (4) used for the solution for y is plotted also on Fig. 28. The graph of equation (4) cuts the y-axis at the values of y corresponding to the y values for the common solution of the two original equations, that is at $y = 0$, and $y = {}^{8}/_{5}$, and the corresponding values for x are obtained by substituting these values of y in equation (1), giving the same common solution $x = {}^{6}/_{5}$, $y = {}^{8}/_{5}$ and $x = 2$, $y = 0$.

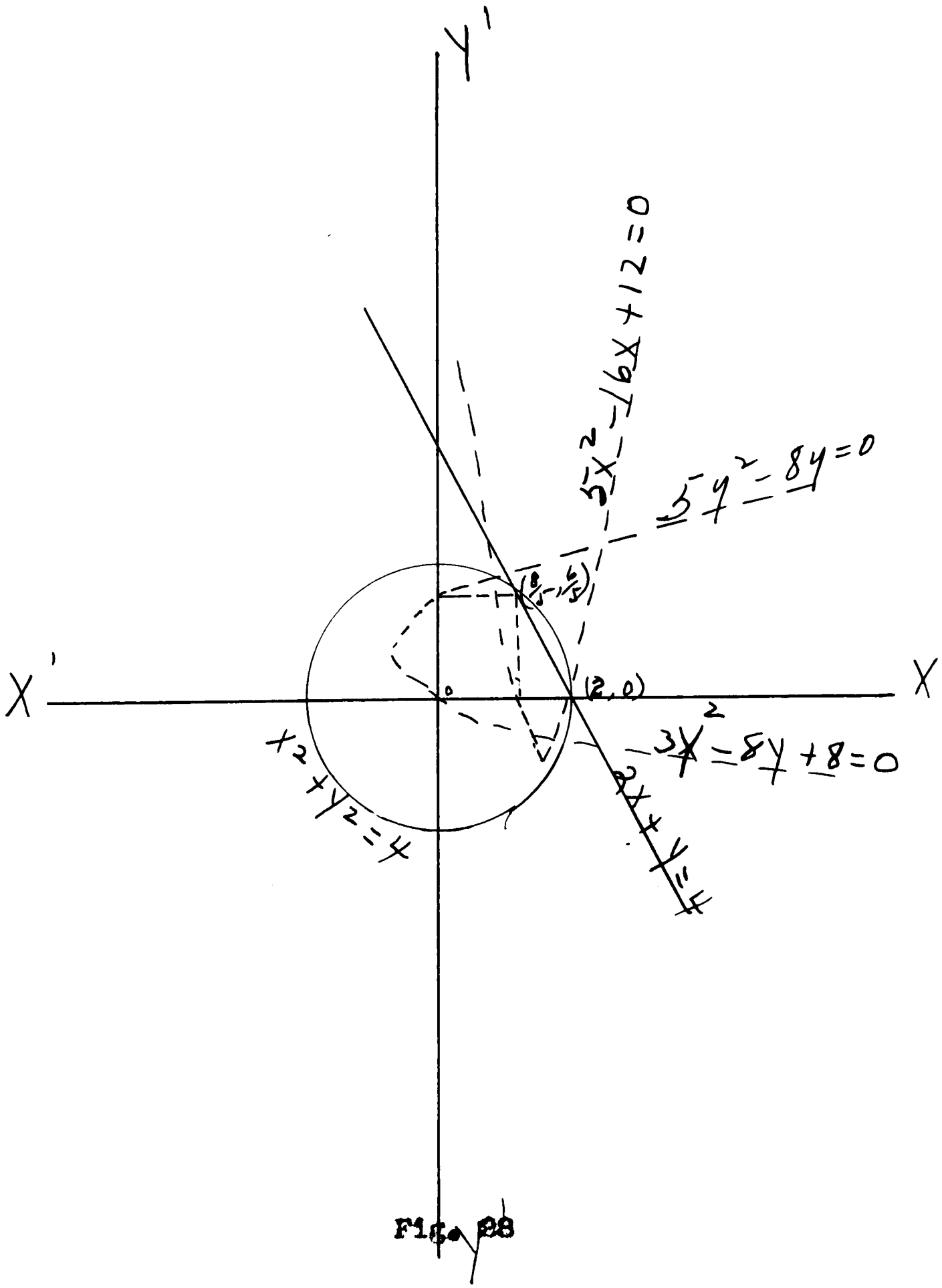


Fig. 28

The next system of equations we will consider is

$$(1) \qquad x^2 + xy + y^2 = 39$$

$$(2) \qquad x + y = -2.$$

Solving (2) for y and substituting in (1)

$$(3) \qquad x^2 + 2x - 35 = 0$$

$$x = 5, -7$$

$$y = -7, 5.$$

The graphical representation is shown in Fig. 29 with the graph of the equation (3) used for the purpose of the solution for x. The graph of equation (3) cuts the x-axis at the values of x corresponding to the x values for the common solution of the two original equations, that is at $x = 5$ and $x = -7$, and the corresponding values for y in the case just mentioned are obtained by the use of these values of x in equation (2), giving the common solution $x = 5$, $y = -7$ and $x = -7$, $y = 5$.

We will consider the equation (1) and (2) as before but getting the auxiliary equation in terms of y.

$$(1) \qquad x^2 + xy + y^2 = 39$$

$$(2) \qquad x + y = -2.$$

Solving (2) for x and substituting in (1)

$$(4) \qquad x^2 + 2y - 35 = 0.$$

$$y = 5, -7$$

$$x = -7, 5.$$

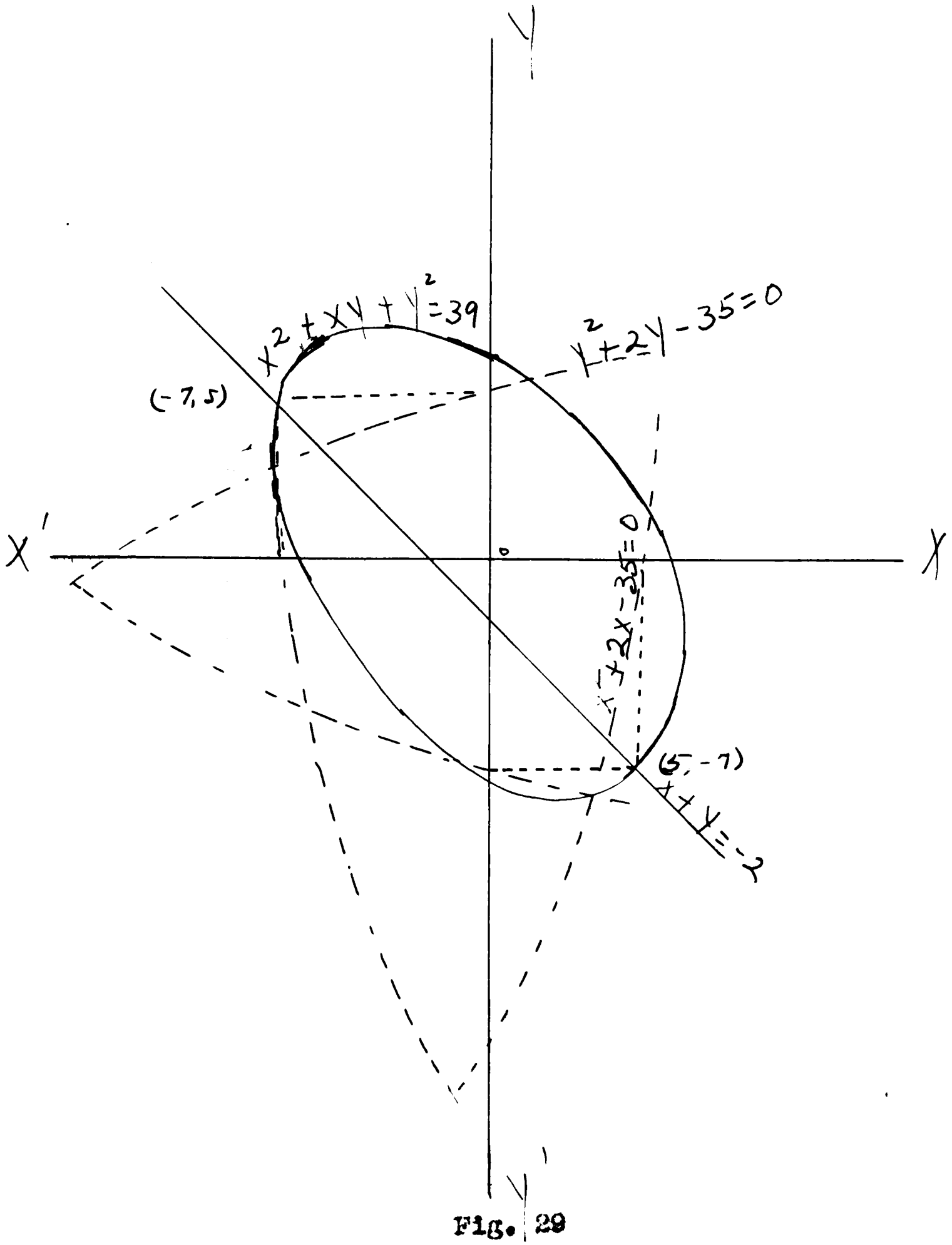


Fig. 29

The graphical representation of equation (4) is shown in Fig. 29. The graph of this equation used for the solution of y cuts the y-axis at the values of y corresponding to the y values of the common solution of the two original equations, $y = 5$ and $y = -7$.

We will consider the system of equations:

$$(1) \qquad x^2 + y^2 = 16$$

$$(2) \qquad 3x^2 - 2y^2 = -8.$$

Solving:

$$x = \pm .4\sqrt{30}.$$

$$y = \pm .4\sqrt{70}.$$

The graphical solution of (1) and (2) is shown on Fig. 30 with the graph of the values, $x = \pm .4\sqrt{30}$ and the graph of the values, $y = \pm .4\sqrt{70}$ plotted which cross the x-axis and y-axis at the values of $x = \pm .4\sqrt{30}$ and $y = \pm .4\sqrt{70}$ and also cuts the graph of the two original equations at their points of intersection, $(.4\sqrt{30}, .4\sqrt{70})$ $(.4\sqrt{30} - .4\sqrt{70})$ $(- .4\sqrt{30}, .4\sqrt{70})$ and $(- .4\sqrt{30}, - .4\sqrt{70})$.

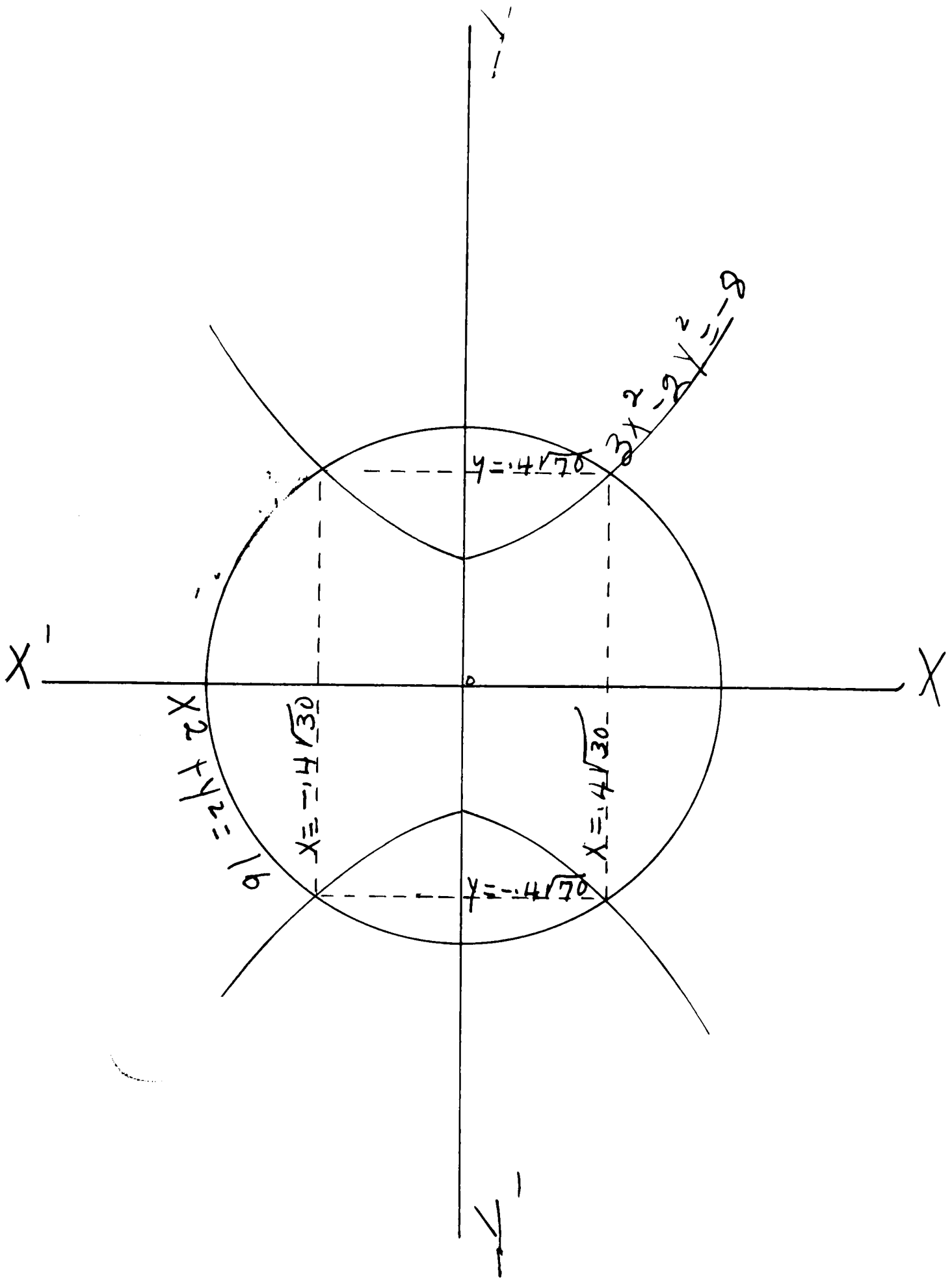


Fig. 30

We will consider the system of equations

$$(1) \quad 4x^2 + y^2 = 16$$

$$(2) \quad y^2 = 9 + 2x.$$

Solving (2) for y^2 and substituting in (1)

$$(3) \quad 4x + 2x - 7 = 0$$

$$x = .1\sqrt{110}, \; y = \pm .2\sqrt{173}$$

$$x = -.4\sqrt{15}, \; y = \pm .2\sqrt{277}.$$

The graphical solution of (1) and (2) is shown on Fig. 31. Equation (3) by which the value of $\underline{x}$ is found is also plotted on Fig. 31. The graph of equation (3) cuts the x-axis at the values of $\underline{x}$ corresponding to the $\underline{x}$ values for the common solution of the two original equations that is $x = .1\sqrt{110}$ and $x = -.4\sqrt{15}$. The corresponding values for $\underline{y}$ are found by use of these values of $\underline{x}$ in (1).

$$x = .1\sqrt{110}, \; y = \pm .2\sqrt{173}$$

$$x = -.4\sqrt{15}, \; y = \pm .2\sqrt{277}.$$

We will consider the same system of equations

$$(1) \quad 4x^2 + y^2 = 16$$

$$(2) \quad y^2 = 2x + 9.$$

Solving (2) for $\underline{x}$ and substituting in (1)

$$y^4 - 17y + 65 = 0$$

$$y^2 = \frac{17 \pm \sqrt{29}}{2}$$

$$y = \pm .2\sqrt{173} \qquad x = .1\sqrt{110}$$

$$y = \pm .2\sqrt{277}. \qquad x = -.4\sqrt{15}.$$

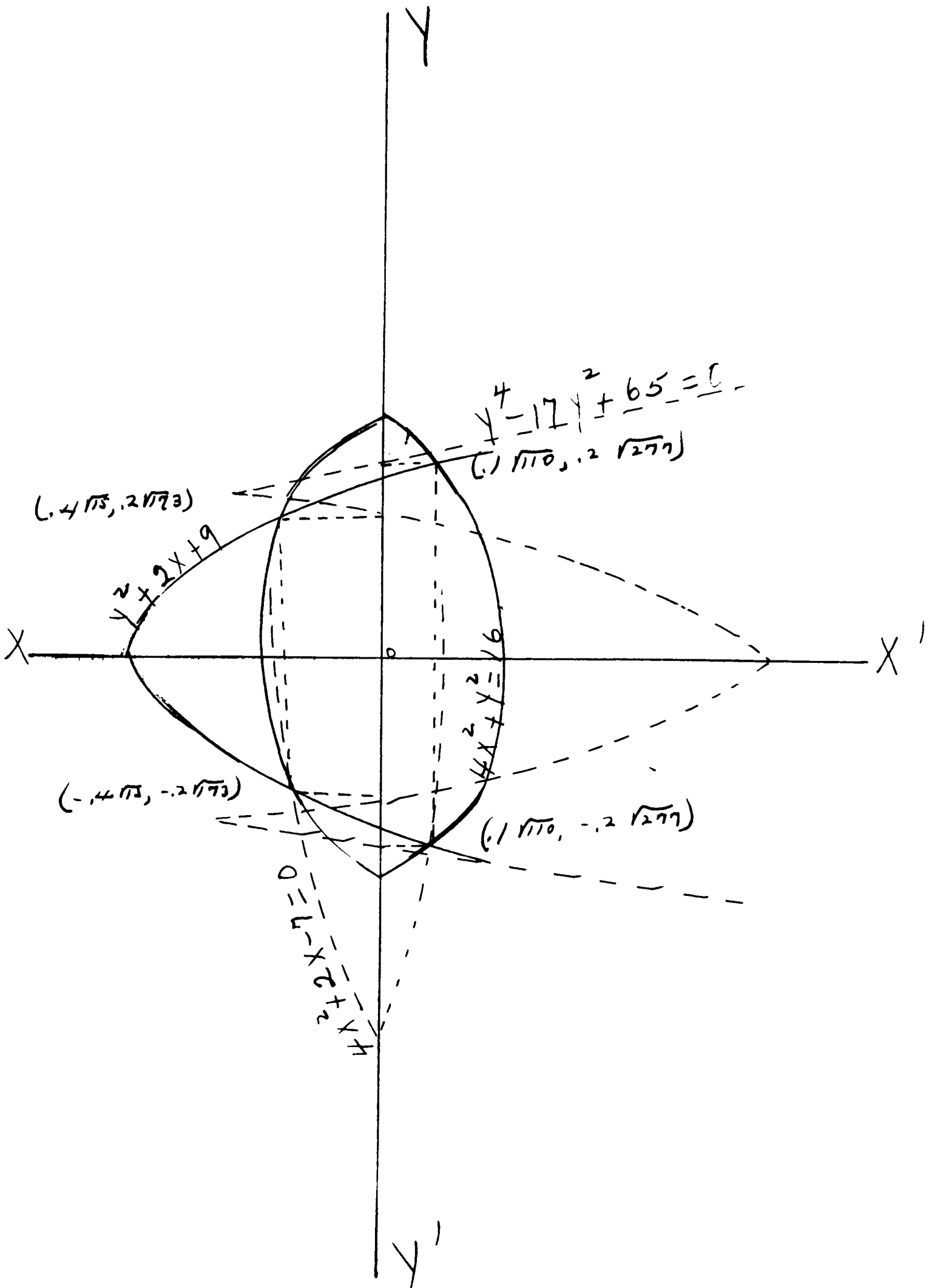

Y
X
X'
Y'
$y^4 - 17y^2 + 65 = 0$
$(.1\sqrt{110}, .2\sqrt{277})$
$(.4\sqrt{15}, .2\sqrt{193})$
$y^2 + 2x + 9$
$(-.4\sqrt{15}, -.2\sqrt{193})$
$(.1\sqrt{110}, -.2\sqrt{277})$
$y^2 - 2x - 4$

Fig. 31

We will consider the system of equations

$$(1)\ xy = -6$$

$$(2)\ 2x - y = 8.$$

Solving (2) for y and substituting in (1)

$$(3)\ x^2 - 4x + 3 = 0.$$

$$x = 1, 3$$

$$y = -6, -2.$$

The graphical representation of (1) and (2) is shown on Fig. 32 with the graph of equation (3) used to obtain the values for x. The graph of equation (3) cuts the x-axis at the values of x corresponding to the x values of the common solution of the two original equations at $x = 1$ and $x = 3$. The corresponding values of y are found by substituting these x values in (1) giving the common solution $x = 1$, $y = -6$ and $x = 3$, $y = -2$.

We will consider the same system.

$$(1)\quad xy = -6$$

$$(2)\quad 2x - y = 8.$$

Solving (1) for x and substituting in (2)

$$(4)\ y^2 + 8y + 12 = 0$$

$$y = -6, -2$$

$$x = 1, 3.$$

The graph of equation (4) is plotted also on Fig. 32. This graph cuts the y-axis at the values of y corresponding to the x values of the common solution of the two original equations at $y = -6$, and $y = -2$.

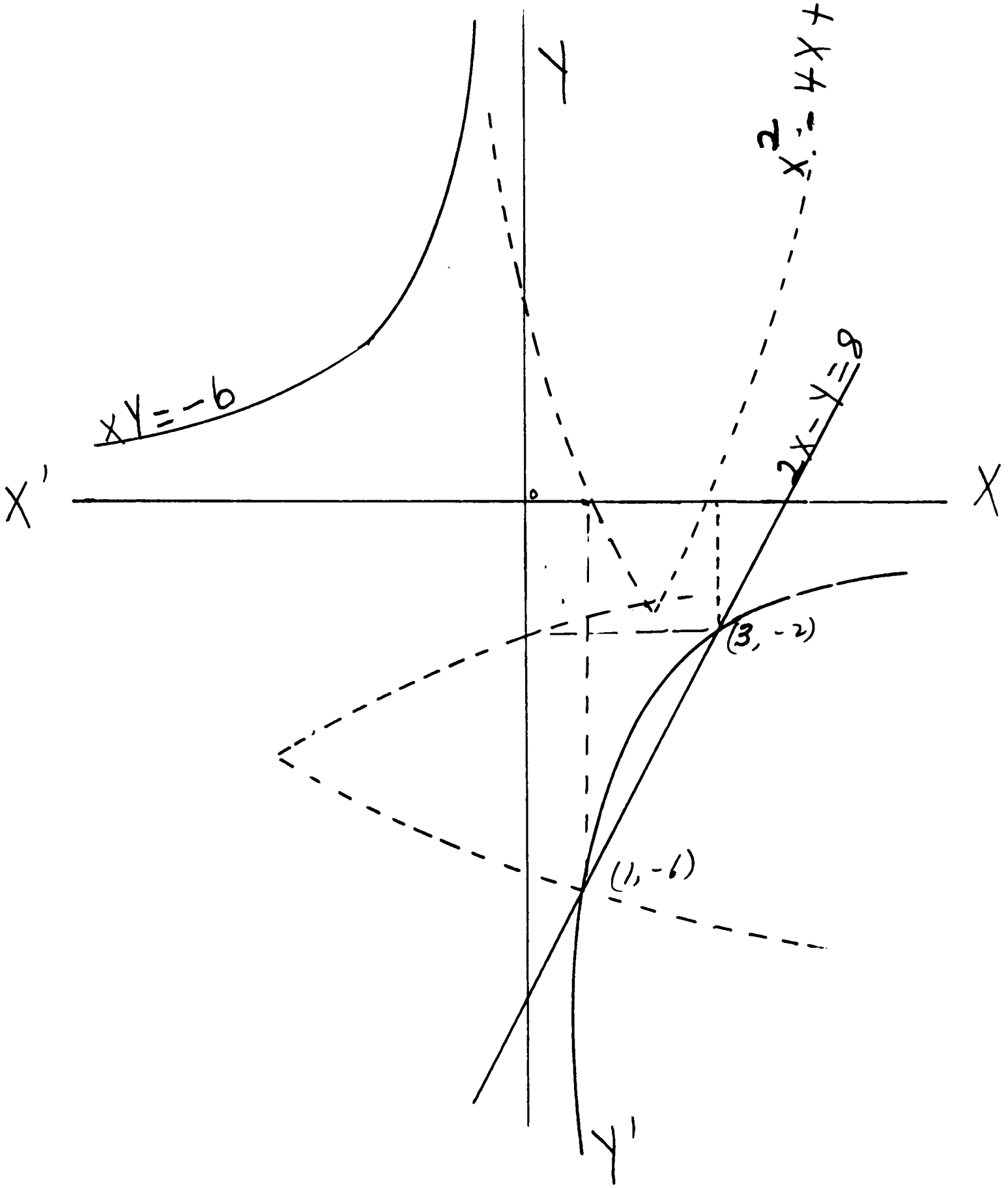


Fig. 32

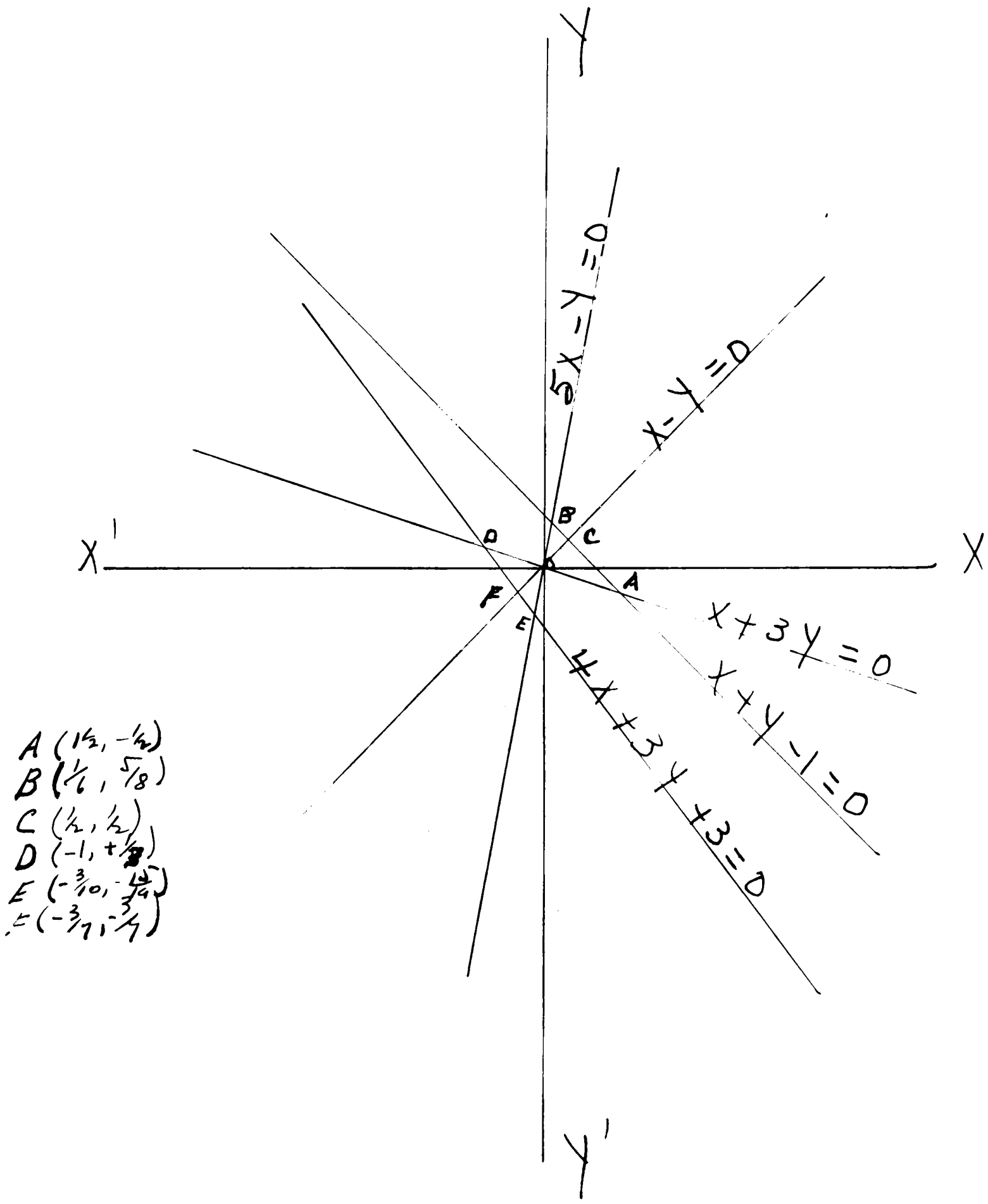

Y
5X - Y = 0
X - Y = 0
B
C
D
X'
X
O
A
F
E
X + 3Y = 0
X + Y - 1 = 0
4X + 3Y + 3 = 0
Y'
A (1/2, -1/2)
B (1/6, 5/8)
C (1/2, 1/2)
D (-1, +1/3)
E (-3/10, -1/4)
F (-3/7, -3/7)

Fig. 33

We will consider the system of equations:

$$(x + 3y)(5x - y)(x - y) = 0$$

$$(x + y + 1)(4x + 3y - 3) = 0.$$

Solving each factor of (1) with each factor of (2) we get:

1. $x + 3y = 0$, $x + y - 1 = 0$
2. $5x - y = 0$, $x + y - 1 = 0$
3. $x + y = 0$, $x + y - 1 = 0$
4. $x + 3y = 0$, $4x + 3y + 3 = 0$
5. $5x - y = 0$, $4x + 3y + 3 = 0$
6. $x - y = 0$, $4x + 3y - 3 = 0.$

The values of x and y are:

1. $x = 1\frac{1}{2}$ $\quad y = -\frac{1}{2}$
2. $x = 1/8$ $\quad y = 5/8$
3. $x = \frac{1}{2}$ $\quad y = \frac{1}{2}$
4. $x = -1$ $\quad y = +1/3$
5. $x = -3/10$ $\quad y = -15/19$
6. $x = -3/17$ $\quad y = -3/7.$

The solution of the equations (1) and (2) is shown graphically on Fig. 33. The graph is a straight line for each linear factor. The values are calculated when one line of a factor of equation (1) crosses a line of a factor of equation (2).

CHAPTER IV

INFINITE SOLUTIONS OF TWO EQUATIONS

This chapter will deal with finite and infinite solutions showing the relations between curves when solutions are infinite.

We will consider the system of equations

$$(1) \qquad y^2 - x^2 + 2x + 2y + 4 = 0$$

$$(2) \qquad y + mx = 0.$$

If we let $m = 0$, then $y = 0$ and the graph of $y + mx = 0$, becomes the x-axis and cuts the curve in two places. We then have two finite solutions.

$$(1) \qquad y^2 - x^2 + 2x + 2y + 4 = 0$$

$$(3) \qquad y + 0x = 0.$$

Solving (1) and (3)

$$-x^2 + 2x + 4 = 0$$

$$x = 1 \pm \sqrt{5}$$

$$y = 0.$$

Let $m = -1$, then the equations become

$$(1)\quad y^2 - x^2 + 2x + 2y + 4 = 0$$

$$y - x = 0$$

Solving (4) for y and substituting in (1)

$$x^2 - x^2 + 2x + 2x + 4 = 0$$

$$0\,x^2 + 4x + 4 = 0$$

The coefficient of the second degree vanishes, thereby giving an infinite solution and one finite solution, $x = -1$, $y = -1$.[1]

Let $m = 1$, then the equations become

$$(1)\quad y^2 - x^2 + 2x + 2y + 4 = 0$$

$$(4)\quad y + x = 0.$$

Solving (4) for y and substituting in (1)

$$x^2 - x^2 + 2x - 2y + 4 = 0$$

$$0\,x^2 + 0\,x + 4 = 0.$$

The coefficients of the first and second degree terms both vanish. Therefore there are no finite solutions but both infinite solutions.[2] These conditions are shown graphically in Fig. 34.

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1. Fine's College Algebra-Page 318.
2. Ibid.

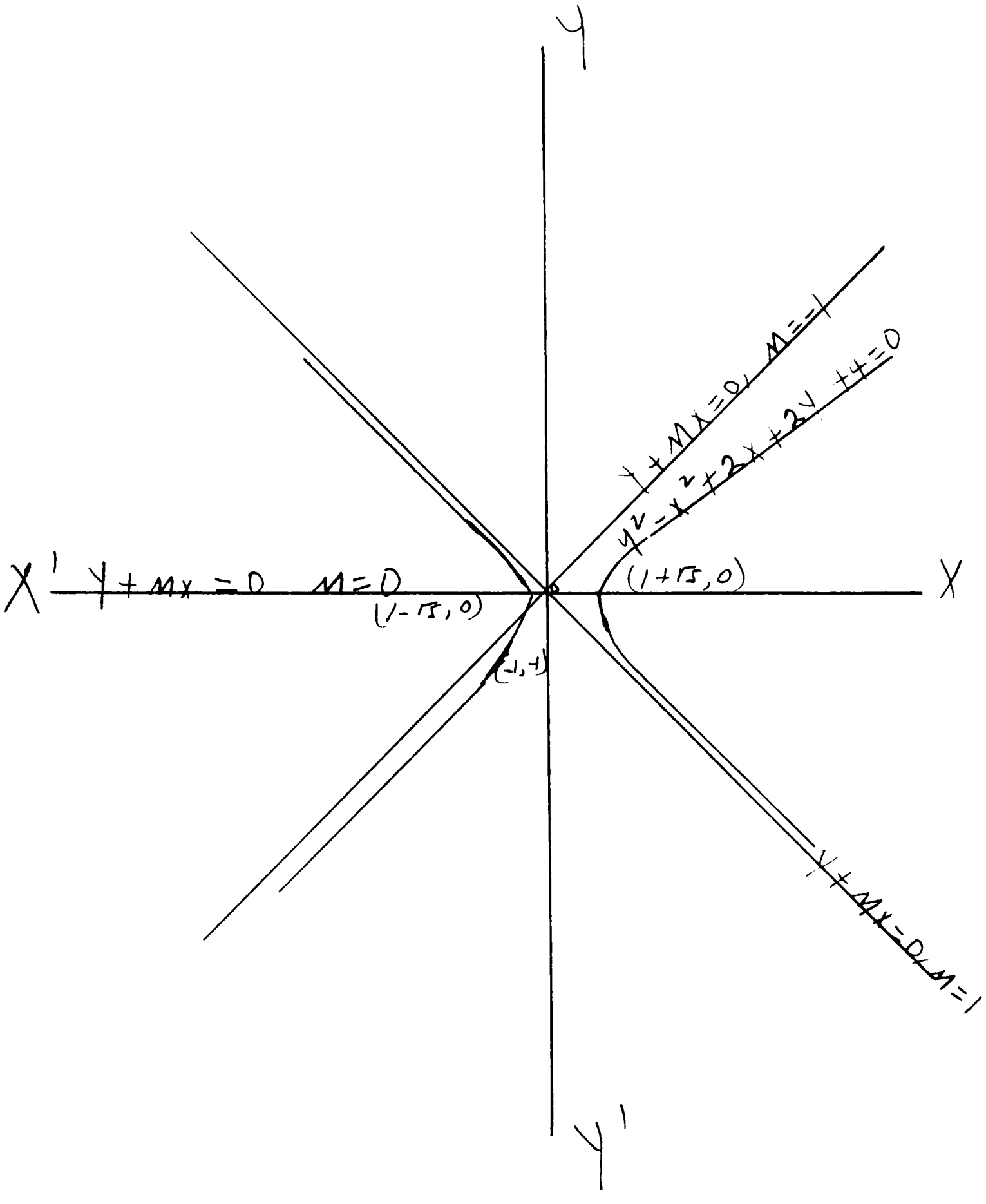

Fig. 34

We will consider another system of equations.

$$(1) \qquad y^3 + 4x + 4 = 0$$

$$(2) \qquad y = mx.$$

If we let m = 2, the equation become

$$(1) \qquad y^2 + 4x + 4 = 0$$

$$(3) \qquad y = 2x.$$

Solving (1) and (3)

$$4x^2 + 4x + 4 = 0$$

$$x = \frac{-1 \pm \sqrt{3}\ i}{2}, \quad (i = \sqrt{-1})$$

$$y = -1 \pm \sqrt{3}\ i.$$

The common solutions are imaginary[1] and the curves do not meet. The common solutions are infinite[2] if the coefficient of the highest degree terms vanishes but if such a coefficient does not vanish the solutions are finite.

Let m = 1, then the equations become

$$(1) \qquad y^2 + 4x + 4 = 0$$

$$(4) \qquad y = x.$$

Solving (1) and (4)

$$x^2 + 4x + 4 = 0$$

$$x = -2, \quad y = -2$$

$$x = -2, \quad y = -2.$$

---

1. Fine's College Algebra-p-296.
2. Fine's College Algebra-p-318.

The two common solutions are coincident and the straight line is tangent to the curve at (-2, -2) and therefore both solutions are finite and identical.

Let $m = \frac{1}{8}$, then the equations become

$$(1) \qquad y^2 + 4x + 4 = 0$$

$$(5) \qquad y = \tfrac{1}{8}x.$$

Solving (1) and (5),

$$x^2 + 16x + 16 = 0$$

$$x = -8 \pm 4\sqrt{3}$$
$$y = -4 \pm 2\sqrt{3}.$$

Both common solutions are finite and the curves meet at the points found in the solution above.

Let $m = .1$, then the equations become

$$(1) \qquad y^2 + 4x + 4 = 0$$

$$(6) \qquad y = .1\,x.$$

Solving (1) and (6),

$$.01\,x^2 + 4x + 4 = 0$$

$$x = -200 \pm 60\sqrt{11}$$
$$y = -20 \pm 6\sqrt{11}.$$

Both solutions are finite and the two graphs intersect twice.

Let $m = 0$, then the equations become

$$(1) \qquad y^2 + 4x + 4 = 0$$

$$(7) \qquad y = 0\,x.$$

Solving (1) and (7)

$$0x^2 + 4x + 4 = 0$$

The coefficient of the second degree term vanishes and therefore there is one infinite solution and one finite solution $x = -1$, $y = 0$.

These conditions are shown graphically on Fig. 35.

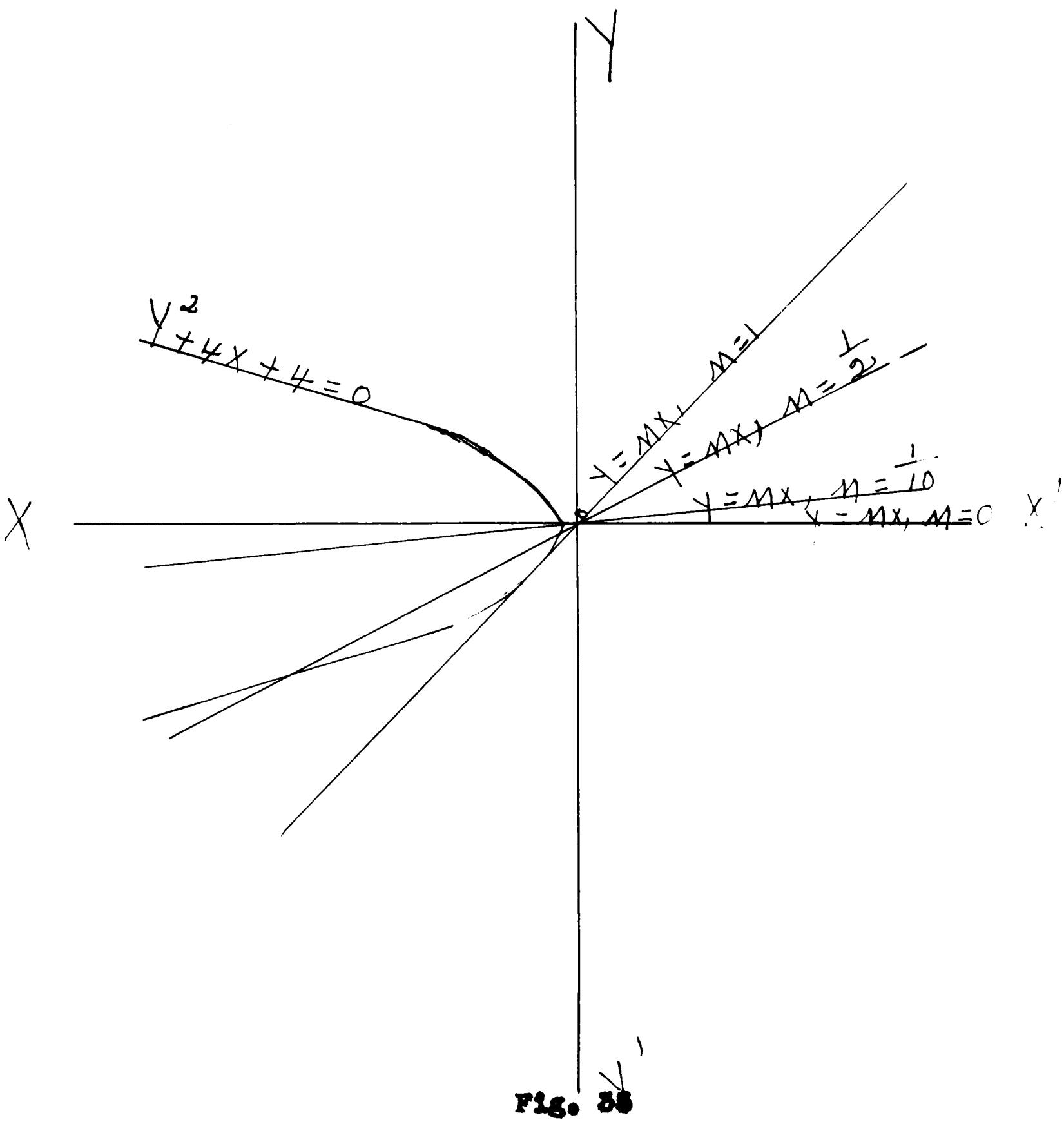

Fig. 35

We will consider the system of equations

$$(1) \qquad 2x^2 + xy - y^2 + 10x + y + 12 = 0$$

$$(2) \qquad 2x - y + 4 = 0$$

Factoring (1)

$$(1) \quad (2x - y + 4)(x + y + 3) = 0$$

$$(2) \qquad 2x - y + 4 = 0.$$

The two equations are plotted on Fig. 36. We find that the second equation will coincide with the graph of the first factor of the first equation and therefore we get and infinite number of finite common solutions. There is one other solution at the point where the graph of the second equation intersects the graph of the second factor of equation (1) $x = -7/3$, $y = -2/3$.

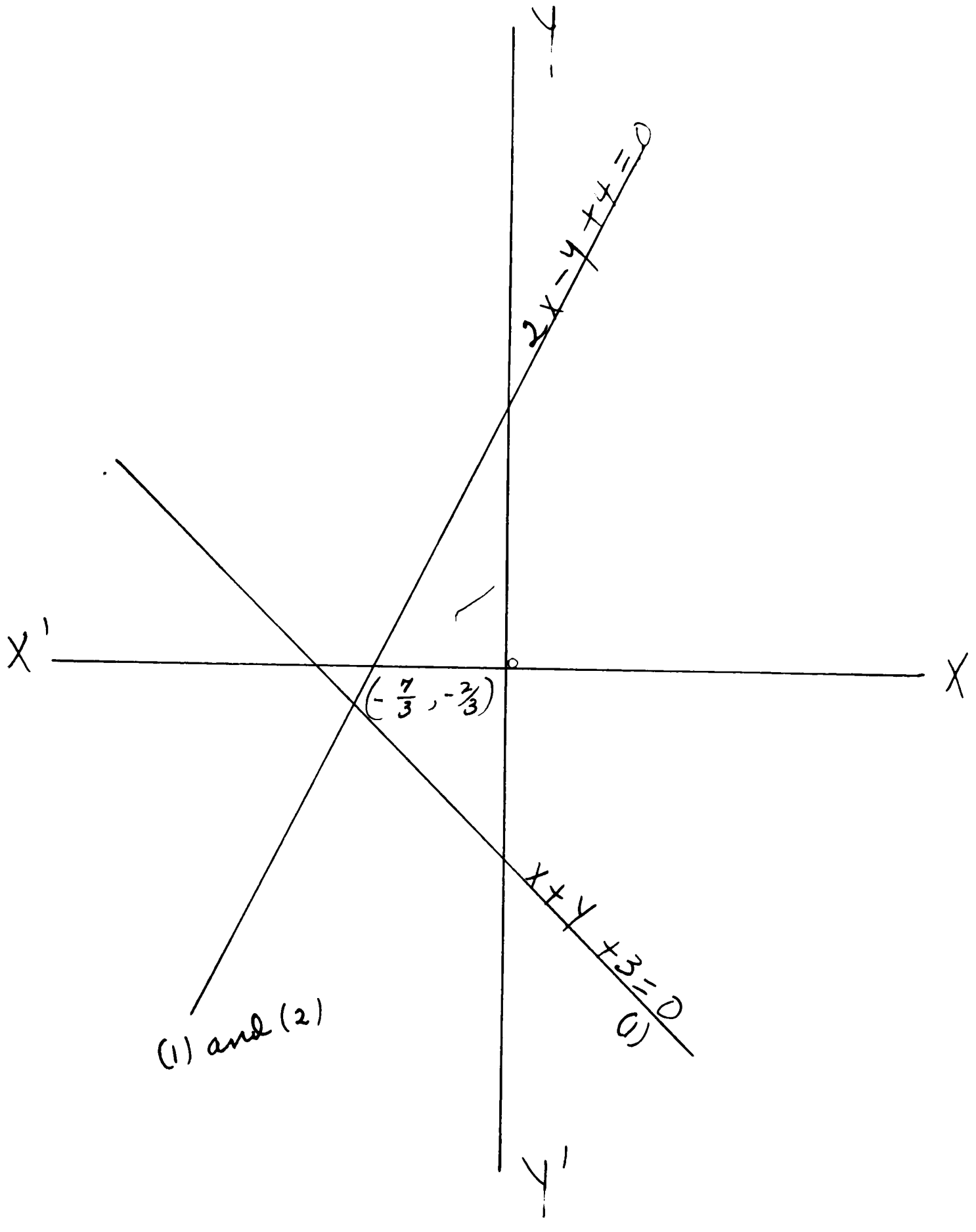

Fig. 33

We will consider now the system of equations:

$$(1) \quad x^3 + y^3 = 27$$

$$(2) \quad x + y = 3.$$

Dividing (1) by (2)

$$(3) \quad x^2 - xy + y^2 = 9$$

Solving (3) with (2) we get two finite solutions,

$$x = 3, \quad y = 0$$

$$x = 0, \quad y = 3.$$

The graphs of (1) and (2) with their common solutions are shown on Fig. 37, with the graph of equation (3) which was used in the obtaining values for x and y. Since equation (3) has the two variables x and y in it, its graph will pass through the intersection of the graphs of (1) and (2), at the points $x = 3$, $y = 0$ and $x = 0$, $y = 3$, which is the same as the values of x and y found in the common solution of the two original equations.

$$(1) \quad x^3 + y^3 = 27$$

$$(2) \quad x + y = 3.$$

Solving for y in (2) and substituting in (1)

$$x^3 + (3 - x)^3 = 27$$

$$x^3 + 27 - 27x + 9x^2 - x^3 = 27.$$

$$(1 - 1)x^3 + 9x^2 - 27x = 0$$

$$0 (x^3) \times 9x^2 - 27x = 0.$$

Since the coefficient of the x^3 is zero, we have the one infinite solution and two finite solutions obtained from $9x^2 - 27x = 0$, or $x = 0, 3$. These two values of $\underline{x}$ substituted into

(2) gives $x = 0, 3$

$y = 3, 0,$

which are the same common solutions obtained by the other method of solving (1) and (2).

The presence of infinite common solutions when systems of equations are solved supply the necessary number of common solutions to complete the total number indicated by the theorem[1] which states that the number of common solutions for two equations is the same as the product of the degrees of the two given equations.

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1. H. B. Fine, College Algebra-p. 519.

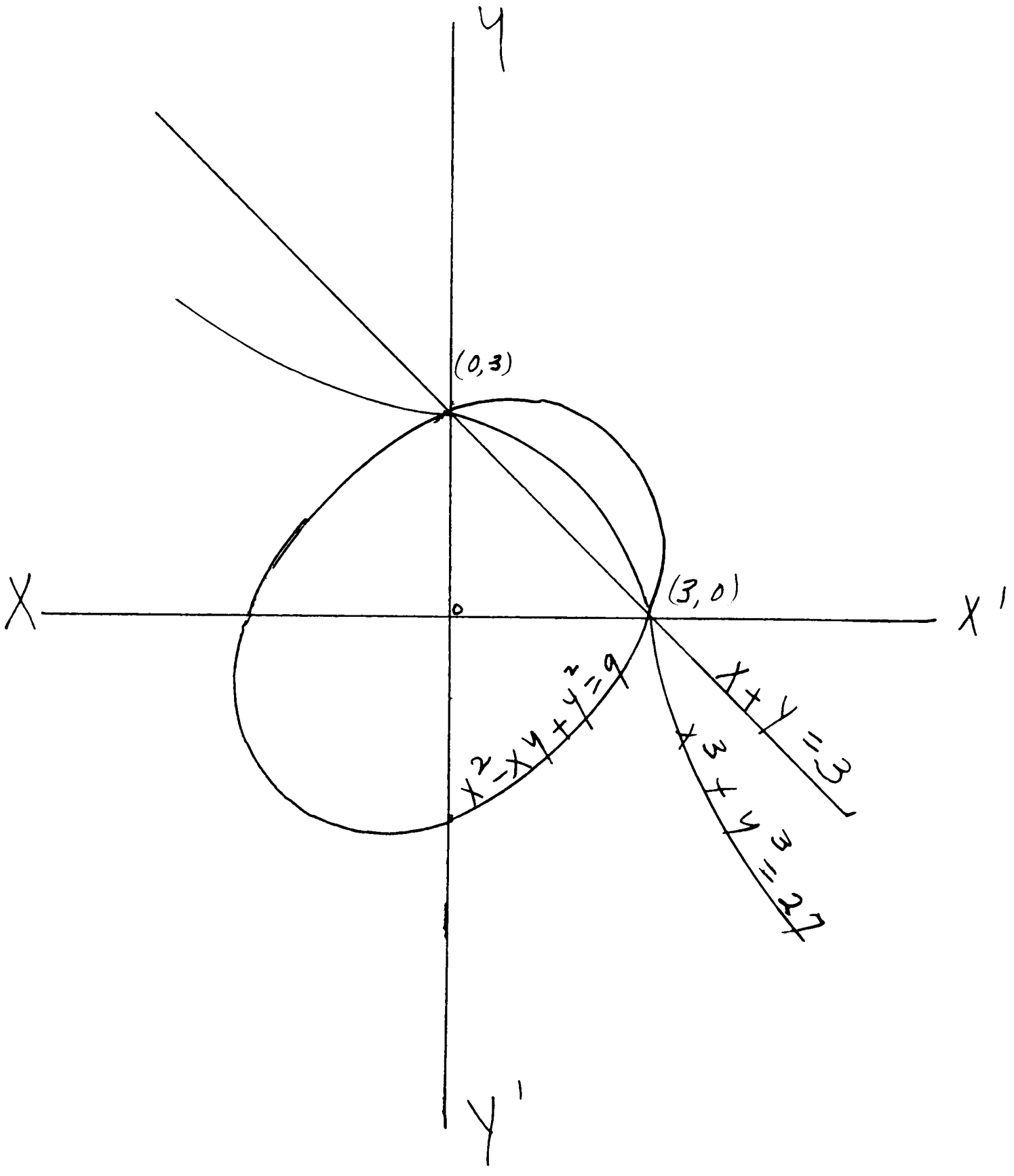

Fig. 37

BIBLIOGRAPHY

1. Edgerton, Edward I. and Carpenter, Perry A., A Second Course in Algebra. Allyn and Bacon, New York, 1934.

2. Fine, Henry Burchard, A College Algebra. Ginn and Company, Boston, 1904.

3. Rietz, H. L. and Crathorne, A. R., Introductory College Algebra. Henry Holt and Company, New York, 1933.

4. Wilson, W. A. and Tracy, J. I., Analytic Geometry. D. C. Heath and Company, New York, 1925.

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